

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS-HME 2023)
Proposed TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. Information Technology					B.E. (SEM : VII)					
Course Name: Machine Learning for IT Application Development					Course Code: PCC- IT 701					
Teaching Scheme (Program Specific)					Examination Scheme (Academic)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)		Practical/Oral (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	IS E	I E	ESE	OR	TW	
3	-	2	5	4	20	20	60	25	25	150
ISE: In-Semester Examination - Paper Duration – 1 Hour ESE: End Semester Examination - Paper Duration - 2 Hours Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: Linear Algebra, Calculus, Probability, Statistics.										

Course Objective:

The course intends to deliver the fundamentals of Machine Learning, in particular focusing on the core concepts of supervised, unsupervised, semi supervised and reinforcement learning. Students will learn the algorithms which underpin many popular Machine Learning techniques, as well as developing an understanding of the hypothetical relationships between these algorithms.

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	Understand basics of Machine Learning	L1, L2, L3	2,3	1,2
2	Apply pre-processing techniques	L1, L2, L3	1,2,4	1,2
3	Understand and apply regression for learning and assess the outcome	L1, L2, L3, L4	1,2,3,4	1,2
4	Apply classification for learning and assess the outcome	L1, L2, L3, L4	1,2,3,4	1,2
5	Apply optimization techniques for performance enhancement	L1, L2, L3, L4	2,4,5	1,2
6	Illustrate semi supervised and reinforcement learning concepts	L1, L2, L3, L4	2,4	1,2

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction to Machine Learning	5	L1, L2
	Machine Learning terminology, Types of Machine Learning, Issues in Machine Learning, Application of Machine Learning, Steps in developing ML application, how to choose the right algorithm, Hypothesis Testing		
2	Data Handling Techniques	6	L1, L2, L3
	Data Handling Techniques, Introduction to Feature selection and extraction, Feature Selection and Extraction Methods, Imbalanced Dataset Handling Techniques		
3	Regression Techniques and Advanced Clustering	10	L1, L2, L3, L4
	Simple and multiple Linear Regression, Logistic Regression, Regularization, Evaluation Metric, Probabilistic Model- Based Clustering, BIRCH, Use cases		
4	Classification Techniques	10	L1, L2, L3, L4
	k Nearest Neighbour, Support Vector Machine, Decision Tree (CART), Issues in Decision Tree, Difference between Regression and classification, Ensembles Techniques -Bagging, Boosting, Evaluation Metrics, Use cases		
5	Optimization Techniques	7	L1, L2, L3, L4
	Model Selection techniques, Cross Validation, Grid Search method, Gradient descent, Types of Gradient descent, Hyper parameter tuning		
6	Semi Supervised Learning & Reinforcement Learning	7	L1, L2, L3, L4
	Semi-supervised Learning, Benefits and limitations, Semi-supervised Learning Techniques Reinforcement Learning: Basics, Elements, Algorithms, Use cases		
Total Hours		45	

List of Experiments:

Practical No.	Type of Experiment	Experiment topic	Hrs	Cognitive levels of attainment as per Bloom's Taxonomy
1	Basic Experiments	Write a code to the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file	2	L1, L2
2		For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training example	2	L1, L2
3	Design Experiments		2	L1, L2, L3
4		Practical on Imbalanced Data Handling Techniques	2	L1, L2, L3
5		Practical on Feature selection and extraction techniques	2	L1, L2, L3
6		Write a program to implement Linear Regression and Multiple Linear Regression	2	L1, L2, L3, L4
7		Write a program to implement Logistic Regression	2	L1, L2, L3, L4
8	Advanced Experiments	Write a program to demonstrate the working of K nearest neighbour	2	L1, L2, L3, L4
9		Write a program to demonstrate the working of Support vector Machine	2	L1, L2, L3, L4
10	Group Activity/ Case study	Implement Q-learning, a fundamental reinforcement learning algorithm	2	L1, L2, L3, L4
11		Case study on Deep Learning	2	L1, L2, L3, L4
		Mini Project	10	L1, L2, L3, L4
Total Hrs.			30	

Books and References:

Sr. No	Title	Authors	Publisher	Edition	Year
1	Machine Learning In Action	Peter Harrington	DreamTech Press	1 st	2012
2	Introduction to Machine Learning	Ethem Alpaydın	MIT Press	4 th	2020
3	Machine Learning	Tom M. Mitchell	McGraw Hill	2 nd	1997
4	Machine Learning An Algorithmic Perspective	Stephen Marsland	CRC Press	2 nd	2011
5	Machine Learning — A Probabilistic Perspective	Kevin P. Murphy	MIT Press	1 st	2012
6	Pattern Recognition and Machine Learning	Christopher M. Bishop	Springer	-	2006
7	Elements of Statistical Learning	Trevor Hastie, Robert Tibshirani, Jerome Friedman	Springer	-	2017

Online References:

Sr. No.	Website Name	URL	Modules Covered
1	www.analyticvidhya.com	https://www.analyticvidhya.com/%20machine%20learning/	M1-M6
2	www.towardsdatascience.com	https://towardsdatascience.com/machine-learning/home	M1-M6
3	www.coursera.org	https://www.coursera.org/learn/machine-learning	M1-M6

Semester-VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS-HME 2023) Proposed

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

BE Information Technology					B.E (SEM: VII)					
Course Name: Big Data Analytics					Course Code: : PCC- IT 702					
Teaching Scheme (Program Specific)					Examination Scheme (Academic)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/Or al/ Presentation (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR	TW	
3	-	2	5	4	20	20	60	25	25	150
ISE: In-Semester Examination - Paper Duration – 1 Hour										
IE: Innovative Examination										
ESE: End Semester Examination - Paper Duration - 2 Hours										
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: : Database Management System, Data Warehouse and Mining, Machine Learning										

Course Objective: The course intends to explore the fundamentals big data analytics and to learn the analytics techniques using various search methods and visualization techniques. Also, to learn the map reduce techniques and various data stream mining techniques.

Course Outcomes: Upon completion of the course students will be able to

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping (Write only Number/s)	PSO Mapping (Write only Number/s)
1	Explain the motivation for big data systems and identify the main sources of Big Data in the real world.	L1, L2	2,3	1,2
2	Demonstrate an ability to use frameworks like Hadoop, NOSQL to efficiently store retrieve and process Big Data for Analytics.	L2,L3	1,2,4	1,2
3	Implement several Data Intensive tasks using the Map Reduce Paradigm.	L4,L5	1,2,3,4	1,2
4	Apply several newer algorithms for Clustering Classifying and finding associations in Big Data	L4,L5	1,2,3,4	1,2
5	Design algorithms to analyze Big data like streams, Web Graphs and Social Media data.	L5,L6	2,4,5	1,2
6	Design and implement successful Recommendation engines for enterprises.	L5,L6	2,4	1,2

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction to Big Data	03	L1,L2
	Introduction to Big Data, BigData characteristics, types of Big Traditional vs. Big Data business approach, Big Data Challenges, Examples of Big Data in Real Life, Big Data Applications, Emerging trends in big data		
2	Introduction to Big Data Frameworks: Hadoop, NOSQL	08	

	What is Hadoop? Core Hadoop Components; Hadoop Ecosystem; Overview of: Apache Spark, Pig, Hive, HBase, Sqoop What is NoSQL? NoSQL data architecture patterns: Key-value stores, Graph stores, Column family (Bigtable) stores, Document stores, Mongo DB.		L2,L3
3	Map Reduce Paradigm MapReduce: The Map Tasks, Grouping by Key, The Reduce Tasks, Combiners, Details of Map Reduce Execution, Coping with Node Failures. Algorithms Using MapReduce: Matrix-Vector Multiplication by MapReduce, Relational-Algebra Operations, Computing Selections ByMapReduce, Computing Projections by MapReduce, Union, Intersection, and Difference by MapReduce, Computing Natural Join by MapReduce, Grouping and Aggregation by MapReduce, Matrix Multiplication. Illustrating use of Map Reduce with use of real-life databases and Applications.	09	L4,L5
4	Mining Big Data Streams Stream-Management System, Examples of Stream Sources, Stream Queries, Issues in Stream Processing. Sampling Data in a Stream: Sampling Techniques. Filtering Streams: The Bloom Filter. Counting Distinct Elements in a Stream: The Count-Distinct Problem, The Flajolet-Martin Algorithm, Combining Estimates, Space Requirements. Counting Ones in a Window: The Cost of Exact Counts, The Datar-Gionis- Indyk- Motwani Algorithm, Query Answering in the DGIM Algorithm.	07	L4,L5
5	Big Data Mining Algorithms Frequent Pattern Mining: Handling Larger Datasets in Main Memory Basic Algorithm of Park, Chen, and Yu. The SON Algorithm and MapReduce. Clustering Algorithms: CURE Algorithm. Canopy Clustering, Clustering with MapReduce Predictive Analytics- Dimension Reduction using PCA, Simple linear regression- Multiple linear regression- Interpretation of regression coefficients. Visualizations - Visual data analysis techniques- interaction techniques – Systems and applications.	09	L5,L6
6	Big Data Analytics Applications Link Analysis: PageRank Definition, Structure of the web, dead ends, Using Page rank in search engine, Efficient computation of Page Rank:		L5, L6

	PageRank Iteration Using Map Reduce, Topic sensitive Page Rank, link Spam, Hubs and Authorities, HITS Algorithm. Mining Social- Network Graphs: Social Networks as Graphs, Types, and Clustering of Social Network Graphs, Direct Discovery Communities, and Counting triangles using Map-Reduce. Recommendation Engines: Model for Recommendation Systems, Content-Based Recommendations, Collaborative Filtering. Case study: Synthetic Data Generation for Healthcare Analytics, Education: Personalized Learning Recommendation System		
	Total Hours	45	

List of Practical's / Experiments:

Experiment No.	Category of Experiment	Practical/ Experiment Topic	Hrs.	RBT Levels Cognitive levels of attainment as per Bloom's Taxonomy
1	Basic Experiments (Based on measurements of properties)	Hadoop HDFS Practical: -HDFS Basics, Hadoop Ecosystem Tools Overview. - Installing Hadoop. - Copying File to Hadoop. - Copy from Hadoop File system and deleting file. - Moving and displaying files in HDFS. - Programming exercises on Hadoop	2	L1, L2, L3
2		Create HIVE Database and Descriptive analytics-basic statistics.	2	L2, L3
3		Experiment on Hadoop Map-Reduce:	4	L3, L4

		-Write a program to implement a word count program using Map Reduce.		
4	Advance Experiments (Based on Volumetric Analysis)	Implementation of Matrix Multiplication algorithm in Map Reduce.	2	L3, L4
5		Data Stream Algorithms: - Implementing DGIM algorithm using any Programming Language	2	L4, L5
6		Data Stream Algorithms: Implement Bloom Filter using any programming language	4	L4, L5
7		Data Stream Algorithms: Implement Flajolet Martin algorithm using any programming language	2	L4, L5
8		Create Dashboard using PowerBI/ Tableau for HealthCare Sector.	2	L4, L5
9		Case Study: One real life large data application	4	L4, L5
10	Design Based Experiments	Mini Project: Design a Big Data Solution using Hadoop/Streaming/Analytics Tools	6	L6
Total			30	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1.	Big Data Analytics	RadhaShankarmani, M Vijayalakshmi	Wiley Publications	3rd	2016
2.	Hadoop: The Definitive Guide	Tom White	O'Reilly Media	4th	2023
3.	Data Science and Big Data Analytics: Discovering, Analyzing,	EMC Education Services	Wiley India	1st	2023

	Visualizing and Presenting Data				
4.	Hadoop in Practice	Alex Holmes	Alex Holmes	2nd	2015
5	Big Data Analytics with R and Hadoop	VigneshPrajapati	Packt Publishing Limited	1st	2016

Online References:

S. No.	Website Name	URL	Modules Covered
1	NPTEL	https://nptel.ac.in/courses/106104189	M1-M6
2	Coursera – Big Data Specialization	https://www.coursera.org/specializations/big-data	M1-M6
3	www.mmds.org	http://www.mmds.org	M1-M6
4	www.guru99.com	https://www.guru99.com/	M1,M2
5	www.edureka.co	https://www.edureka.co/blog/hadoop-tutorial/	M1, M2
6	www.tutorialride.com	https://www.tutorialride.com/big-data-analytics	M1-M6

B.E. Semester –VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS-HME2023)

TCET Autonomy scheme with effect from A.Y. 2025-26

B.E (Information Technology & Computer Engineering)								SEM: VII			
Course Name: Information Security								Course Code: PEC- IT 7012			
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)						
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation						
Hours Per Week					Theory (100)			Practical/Ora l (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contac t Hours	Credits	ISE	IE	ESE	PR	TW	150	
3	-	2	5	4	20	20	60	25	25		
ISE: In-Semester Examination - Paper Duration – 1 Hours											
IE: Innovative Examination											
ESE: End Semester Examination - Paper Duration - 2 Hours											
The weightage of marks for continuous evaluation of Term work/ Report: Formative (40%), Timely completion of practical (40%) and Attendance/Learning Attitude (20%)											
Prerequisite: Concepts of Cryptography, Network Security, Information Theory & Coding, Computer Networks											

Course Objectives:

- The course intends to deliver the fundamental principles/concepts of Cryptography, Network Security real- time concepts their applications and to apply the knowledge of computer networks for analyzing various algorithms spread over various layer of OSI reference model.
- The course also intends to deliver the fundamentals of encryption techniques, and cryptographic algorithms including secret key cryptography, hashes and message digests, public key algorithms, authentication protocols, and PKI standards, and apply them to techniques such as Kerberos, IPsec, and SSL/TLS and email, analyze cryptographic utilities, authentication mechanisms to design secure applications.

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy
1	Identify information security goals, and classical encryption techniques and acquire fundamental knowledge of the concepts of Symmetric cipher models.	L1, L2, L3
2	Understand, compare, and apply different encryption and decryption techniques to solve problems is related to confidentiality and authentication	L1, L2, L3
3	Apply the knowledge of cryptographic checksums and different digital signature algorithms to achieve authentication and create secure applications.	L1, L2, L3,L4
4	Understand Secure Programs, Program Errors, and Other Malicious Code and identify Objects to be Protected, and Use of Passwords for — Additional Authentication Information.	L1, L2, L3
5	Apply network security basics, analyze different attacks on networks, and evaluate the performance of firewalls and security protocols like SSL, IPSec, and PGP.	L1, L2, L3, L4
6	Apply the knowledge of cryptographic utilities and authentication mechanisms to design secure application	L1, L2, L3,L4

Detailed Syllabus (Total No. of Hours: 45):

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction & Classical Cryptography	05	L1, L2,L3
	Introduction and Classical Cryptography: Security Goals, Services, Mechanisms and attacks, Classical Encryption techniques (Symmetric cipher model, mono-alphabetic and poly-alphabetic substitution techniques: Vignere cipher, Playfair cipher, Affine cipher, Hill cipher, transposition techniques: keyed and keyless transposition ciphers, steganography).		
2	Block Ciphers & Public Key Cryptography	08	L1, L2,L3
	Block Ciphers and Public Key Cryptography: Block cipher principles, block cipher modes of operation, DES, Double DES, Triple DES, Advanced Encryption Standard (AES), Blowfish Public key cryptography: Principles of public key cryptosystems-The RSA algorithm, Diffie Hellman Key exchange algorithm.		
	Cryptographic Hashing & Digital Signatures	07	L1, L2, L3,L4

3	Cryptographic Hashes and Digital Signatures: Authentication requirement – Authentication function, Types of Authentication, MAC – Hash function – Security of hash function and MAC MD5 Message-Digest Algorithm, Secure Hash Algorithm, Digital signature, and authentication protocols: Needham Schroeder Authentication protocol, Digital Signature Schemes – RSA, Digital Certificate: X.509, PKI		
4	Protection of Computing Resources & Security Features	08	L1, L2, L3, L4
	Protection of Computing Resources and Security Features: Secure Programs Non-malicious Program Errors – Buffer Overflows, Incomplete Mediation; Viruses and Other Malicious Code – Methods of Control – Developmental Controls, Objects to be Protected; User Authentication – Use of Passwords, Additional Authentication Information, Attacks on Passwords, Exhaustive Attack, and Password Selection Criteria.		
	Network Security and Application		

5	Network Security: Network Security basics: TCP/IP vulnerabilities (Layer wise), Packet Sniffing, ARP spoofing, port scanning, IP spoofing, TCP syn flood, DNS Spoofing. Denial of Service: Classic DOS attacks, Source Address spoofing, ICMP flood, SYN flood, UDP flood, Distributed Denial of Service, and Defenses against Denial of Service Attacks. Firewalls, Intrusion Detection Systems: Host-Based and Network Based IDS, Honey pots. Network Security applications: Authentication Applications, Kerberos, Internet Security Protocols: SSL, TLS, IPSEC: AH, ESP, Secure Email: PGP and S/MIME, Key Management.	10	L1, L2, L3, L4
6	OWASP top 10 Attacks OWASP top 10: Open web application project OWASP Top 10 attacks : 2013, Transition from 2013 to 2017 and 2017 to 2021 SQL injection, Cross site scripting, Cross Site Request Forgery, Server Side Request Forgery	07	L1, L2, L3,L4



TCET
DEPARTMENT OF INFORMATION TECHNOLOGY (IT)
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education
Under Autonomy - CBCGS-HME 2023
University of Mumbai



List of Practical's/ Tutorials:

Exp No.	Type/Category of Experiment	Practical/ Experiment Topic	Hr s.	Cognitive levels of attainment as per Bloom's Taxonomy
1.	Basic Experiments (Based on measurements of properties)	To implement conventional cryptographic techniques Ceaser Cipher, and Vernam cipher.(Substitution cipher)	2	L1, L2
2.		Study the use of network reconnaissance tools like WHOIS, dig, and traceroute ns lookup to gather information about networks and domain registrars		L1, L2, L3
3.		Implementation of Diffie Hellman Key exchange algorithm	2	L1, L2, L3
4.	Design Based Experiments	Implementation and analysis of DES cryptosystem	2	L1, L2, L3
5.		Demonstrate and test the integrity of the message using MD-5, SHA- a) For varying message size and analyze the performance of the two protocols. Use crypt APIs.	2	L1, L2, L3
6.		Implementation and analysis of RSA cryptosystem and Digital signature scheme using RSA/El Gamal	2	L1, L2, L3, L4
7.		Study of packet sniffer tools Wireshark, :- a). Observer performance in promiscuous as well as non-promiscuous mode. b) Show the packets can be traced based on different filters	2	L1, L2, L3
8.		Demonstrate the use of nmap with different options to scan open ports, perform OS fingerprinting, ping scan, tcp port scan, udp port scan, etc.	2	L1, L2, L3
9.		Study and Simulation of DOS attack using Hping and other tools		L1, L2, L3, L4
10.		Study and use the NESSUS/ISO Kali Linux tool to scan the network for vulnerabilities	2	L1, L2, L3
11.	Advance Experiments (Based on)	Set up IPSEC under LINUX and explore the GPG tool of Linux to implement email security	4	L1, L2, L3, L4

12.	Volumetric Analysis)	Study and perform CSRF and SSRF Attack using Burpsuite	2	L1, L2, L3, L4
13.		Study and perform XSS Attack using Burp suite	2	L1, L2, L3, L4
14.	Project Based Experiments- Group Activity) (Students should complete any one project Based experiment from the list or any other project in discussion with Faculty in-Charge)	Case study , Mini Project	6	L1,L2,L3,L4, L5
Total			30	

Books and References:

S. No.	Title	Authors	Publisher	Edition	Year
1	Information Security Principles and Practice	Mark Stamp, Deven Shah	Cengage Learning	2nd Edition	2011
2	Cryptography & Network Security	Behrouz A. Ferouzan	Tata Mc Graw Hill	2nd Edition	2008
3	Cryptography and Network Security, Principles and Practice	William Stallings	Pearson Education	6 th Edition	2013
4	Cryptography & Network Security	Bernard Menezes	Cengage Learning	1 st Edition	2010
5	Cryptography and Network Security	Atul Kahate	Tata Mc Graw Hill	2nd Edition	2008
6	Security in Computing	Charles P. Pfleeger	Pearson Education	5 th Edition	2015
7	Information System Security	Nina Godbole	Wiley	2nd Edition	2017

Online References:

S. No.	URL
1	https://nptel.ac.in/courses/106105031/
2	https://www.coursebuffet.com/course/814/nptel/cryptography-and-network-security-iit-kharagpur
3	OWASP TOP 10: https://www.owasp.org/index.php/Top_10_2013

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic & Multidisciplinary Education (CBCGS-HME 2023)
Proposed TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

BE Information Technology					B.E (SEM: VII)					
Course Name: DevOps					Course Code: : PEC-IT 7013					
Teaching Scheme (Program Specific)					Examination Scheme (Academic)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/Oral/ Presentation (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR	TW	150
3	-	2@	5	4	20	20	60	25	25	
ISE: In-Semester Examination - Paper Duration – 1 Hour										
IE: Innovative Examination										
ESE: End Semester Examination - Paper Duration - 2 Hours										
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: : Operating System, Virtualization, Cloud Computing, Java and Web Programming, and Software Engineering.										

Course Objective: This course introduces students to the core principles and practices of DevOps, focusing on automation, collaboration, and agile methodologies. It covers tools like Git, Jenkins, Docker, and Ansible to implement CI/CD, configuration management, and monitoring.

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	CO1: Describe the evolution, objectives, principles, and challenges of DevOps in modern software development.	L1, L2	1, 2, 5, 11	1
2	CO2: Explain and illustrate DevOps automation practices such as version control, continuous integration, and tool pipelining.	L1, L2, L3	1, 3, 5, 11	1, 2
3	CO3: Apply and manage Git-based version control workflows including setup, branching, merging, and working with remote repositories.	L1, L2, L3, L4	1, 3, 5, 8	1, 2

4	CO4: Analyse and evaluate CI/CD concepts, tools, and practices using Jenkins for building and deploying applications.	L1, L2, L3, L4	1, 2, 3, 4, 5	1, 2
5	CO5: Build and deploy containerized applications using Docker, and compare containerization with virtualization.	L1, L2, L3, L4, L5	1, 3, 5, 6	1, 2
6	CO6: Design and implement continuous testing and configuration management using tools like Selenium, Ansible, Chef, or Puppet.	L1,L2,L3,L4,L5,L6	1, 3, 5, 7, 10, 11	1, 2, 3

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction to DevOps	7	L1,L2
	What Is DevOps, History of DevOps, DevOps Objectives, DevOps Principles, DevOps and Software Development Life Cycle, Waterfall Model, Agile Model, DevOps Delivery Pipeline, Market Trend of DevOps, DevOps Technical Challenges, DevOps Ecosystem, DevOps Engineer Skills in the market, & DevOps on the Cloud.		
2	DevOps and Automation	8	L1,L2
	Version Control, Continuous Integration, Continuous Testing, Configuration Management, Continuous Deployment, Containerization, Continuous Monitoring, Tool pipelining		
3	Version Control Systems	8	L1,L2,L3,L4,L5,L6
	Version control systems , Version Control System types, Difference between CVCS and DVCS, Introduction to GIT, Essentials of GIT in industry, How to Setup GIT, Common commands in GIT, Working with Remote Repositories, Branching and Merging in GIT, GIT workflows, GIT cheat sheet, other version controls, CVS , Mercurial		
4	Continuous Integration and Continuous Deployment (CI/CD)	8	L1,L2,L3,L4,L5,L6
	Introduction to Continuous Integration, Continuous Delivery and Deployment, CI/CD principles and their importance, Benefits of CI/CD, continuous integrations tools. Jenkins and its Architecture, Jenkins Management, Build Setup, GIt and Jenkins Integration, Build & Test Applications with Continuous Integration, Scheduling build Jobs		
5	Virtualization & Containerization	8	L1,L2,L3,L4,L5,L6

	Introduction to Container, Understanding containerization and its benefits, Container Life Cycle. Docker Essentials: Learning Docker fundamentals, containers, and images, Use case of Docker, Platforms for Docker, Dockers vs. Virtualization, Understanding images and containers. Installing and Configure Docker for creating Containers of Operating Systems ,Build, deploy and manage web or Java application on Docker, container environment using a Dockerfile.		
6	Continuous Testing and Configuration Management	6	L1,L2,L3,L4,L5
	Introduction to Continuous Testing, Agile Testing Techniques, Life Cycle, testing tools, testing using Selenium tool. Software Configuration Management, provisioning, Understanding tools like Ansible/ Chef/ puppet.		
Total Hours		45	

Capstone Project Guide Lines

1. The mini project work is to be conducted by a group of three students
2. Each group will be associated with a subject Incharge/ mini project mentor. The group should meet with the concerned faculty during Laboratory hours and the progress of work discussed must be documented.
3. The students may do survey for different application which they can apply DevOps.
4. Students will do Installation, configuration and also latest DevOps tools for project
5. Students will try to consider following points in their Mini Project
 - a) Version control.
 - b) Integration tool
 - c) Testing Tool
 - d) Deployment
 - e) Configuration
6. Each group along with the concerned faculty shall identify a potential problem statement for project development, on which the student and implementation is to be conducted.
7. Each group may present their work in various project competitions and paper presentations.
8. A detailed report is to be prepared as per guidelines given by the concerned faculty.

Capstone Project Hours Distribution:

Sr. No.	Work to be done	No. of Hours	Cognitive levels of attainment as per Bloom's Taxonomy

1	Study DevOps Research papers , articles, mini project title Identification	4	L1, L2
2	Project Title finalization and development of Modules	2	L1, L2
3	Design methodology and tools for implementation	4	L1, L2
4	Implementation of Modules phase 1	4	L1, L2, L3
5	Result Phase I	2	L1, L2, L3, L4
6	Implementation of Modules Phase 2	4	L1, L2, L3
7	Result Phase II	2	L1, L2, L3, L4
8	Testing	2	L1, L2, L3, L4
9	Result validation	2	L1, L2, L3, L4, L5
10	Report Writing	4	L1, L2
	Total Hours	30	

Books and References:

Sr. No	Title	Authors	Publisher	Edition	Year
1	The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations	Gene Kim, Patrick Debois, John Willis, Jez Humble	IT Revolution Press	First Edition	2016
2	Accelerate: The Science of Lean Software and DevOps	Nicole Forsgren, Jez Humble, Gene Kim	IT Revolution Press	First Edition	2018
3	Learning DevOps: Continuously Deliver Better Software	Mikael Krief	Packt Publishing	First Edition	2019
4	Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling at Scale	Jennifer Davis, Katherine Daniels	O'Reilly Media	First Edition	2016
5	Docker Deep Dive	Nigel Poulton	Leanpub	3rd Edition	2018
6	Version Control with Git	Jon Loeliger, Matthew McCullough	O'Reilly Media	2nd Edition	2012

7	Jenkins: The Definitive Guide	John Ferguson Smart	O'Reilly Media	First Edition	2011
8	Ansible for DevOps	Jeff Geerling	Leanpub	2nd Edition	2020

Online References:

Sr. No.	Website Name	URL	Modules Covered
1	DevOps Official Site	https://www.devops.com	M1, M2, M3, M4, M5, M6
2	Atlassian Git Tutorials	https://www.atlassian.com/git	M3
3	Docker Documentation	https://docs.docker.com	M5
4	Jenkins Documentation	https://www.jenkins.io/doc	M4
5	GitHub Learning Lab	https://lab.github.com	M3
6	Selenium Official Site	https://www.selenium.dev	M6
7	Ansible Documentation	https://docs.ansible.com	M6
8	Puppet Documentation	https://puppet.com/docs	M6
9	Chef Documentation	https://docs.chef.io	M6
10	DevOps Tutorials (Edureka)	https://www.edureka.co/blog/devops-tutorial	M1, M2

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic & Multidisciplinary Education (CBCGS-HME 2023)
Proposed TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. (Information Technology)							B.E. (SEM: VII)		
Course Name: Mobile Application Development							Course Code: PEC- IT 7014		
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)				
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation				
Hours Per Week					Theory (100)		Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	IA	ESE	OR	TW	150
3	-	2@	5	4	40	60	25	25	
IA: In-Semester Assessment- Paper Duration – 1Hours									
ESE: End Semester Examination- Paper Duration - 3 Hours									
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Mini Project and presentation (40%) and Attendance /Learning Attitude (20%). @ Capstone Project									
Prerequisite: JAVA Programming, Internet Programming									

Course Objective: The course intends to deliver the fundamental knowledge of Android platform and its architecture, apply and create Android UI designing, broadcast receivers, Internet services, SQLite Database, integrate multimedia, camera, Location based services and know about Mobile security issues.

Course Outcomes: Upon completion of the course student will be able to:

S. No	Course Outcomes	Cognitive levels of attainment as per bloom's Taxonomy
1	Describe Android platform, Architecture and features.	L1, L2, L3, L4
2	Design User Interface and develop activity for Android App.	L1, L2, L3, L4, L6

3	Use Intent, Broadcast receivers and Internet services in Android App.	L1, L2, L3, L4, L6
4	Design and implement Database Application and Content providers.	L1, L2, L3, L4, L6
5	Use multimedia, camera and Location based services in Android App.	L1, L2, L3, L4, L6
6	Discuss various security issues in Android platform	L1, L2, L3, L6

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction to Android and Architecture of Android	07	L1, L2, L3, L4
	Introduction of Android platform, Android features, Android Marketplace, Evolution of Android OS, Android Application Architecture, Android Development Tools, First Android application, How to run and debug applications (Emulator vs. Real device), Android project structure, XML files, Enhancing the first app		
2	Applications, Activities and Building User Interface	08	L1, L2, L3, L4, L6
	Application: Application Manifest File, Externalizing Resources, Android Application Lifecycle and Android Application Class. User Interface: Fundamental Android UI Design, Layouts, Toast, Button, Toggle Button, Switch Button, Image Button, CheckBox, RadioBox, AlertDialog, Spinner, Auto Complete TextView, RatingBar, DatePicker, TimePicker, ProgressBar, File Download, Introduction to fragments Fragment, Fragment Example.		

	Android Activity: Activity Lifecycle and Android Activity classes, Activity Example, Lifecycle, Creating new views, widget toolbox, Adapters(ArrayAdapters, BaseAdapters)		
3	Intents, Broad Cast receiver and Internet Resources	08	L1, L2, L3, L4, L6
	Intents, Types of Intents (Implicit and Explicit), Linking Activities Using intents, Calling Built-in Applications Using intents, displaying notifications, Creating Intent Filters, Broadcast Receivers, Downloading and Parsing Internet Resources, Using the Download Manager, Internet Services, connecting to Google App Engine, Downloading Data Without Draining the Battery.		
4	Data Persistence and Content Providers	08	L1, L2, L3, L4, L6
	Content Providers: Introducing Android Databases, Introducing SQLiteDatabase, DML & DDL Queries in brief, SQLiteOpenHelper, Cursor, SQLite Programming, Android Debug Bridge(adb) tool, Parsing an XML document, Parsing JSON data, Creating Content Providers, Using Content Providers. Adding Search to Your Application, Native Android Content Providers.		
5	Audio, Video, Camera, Maps, Geocoding and Location Based Services	08	L1, L2, L3, L4, L6
	Playing Audio and Video, Manipulating Raw Audio, Using Audio, Using the Camera for Taking Pictures, Recording Video, Using Media Effects, Adding Media to the Media Store. Using Location-Based Services, Using the Emulator with Location-Based Services, selecting a Location Provider, Finding Your Current Location, Location Updates, Proximity Alerts, Geocoder, Map-Based Activities, Displaying Maps.		
6	Securing and Publishing Android Application	06	L1, L2, L3, L6
	Android Security Model, Android's Manifest Permissions, Mobile Security Issues, Recent Android Attacks, Pen Testing Android. Preparing for Publishing, Deploying APK Files.		
	Total Hr.	45	

Capstone Project Guide Lines

1. The mini project work is to be conducted by a group of three students
2. Each group will be associated with a subject Incharge/ mini project mentor. The group should meet with the concerned faculty during Laboratory hours and the progress of work discussed must be documented.
3. The students may do survey for different application which they can create Apps using Android.
4. Students will do Installation, configuration of Android Studio & to create AVD and also try for Cross platform Integrated Development Environment (Any Open Source Tool).
5. Students will try to Design and implement following points in their Mini Project (Android Apps)
 - a) Widget box for Android phone.
 - b) Use Layouts
 - c) Use Intents
 - d) Use Activity
 - e) Use SQLite
 - f) Use Camera
 - g) Use Location API
 - h) Generate APK file
6. Each group along with the concerned faculty shall identify a potential problem statement for Apps development, on which the study and implementation is to be conducted.
7. Each group may present their work in various project competitions and paper presentations.
8. A detailed report is to be prepared as per guidelines given by the concerned faculty.

Capstone Project Hours Distribution:

Sr. No.	Work to be done	No. of Hours	Cognitive levels of attainment as per Bloom's Taxonomy
1	Study Research papers, articles, mini project title Identification	4	L1, L2

2	Project Title finalization and development of Modules	2	L1, L2
3	Design methodology and tools for implementation	4	L1, L2
4	Implementation of Modules phase 1	4	L1, L2, L3
5	Result Phase I	2	L1, L2, L3, L4
6	Implementation of Modules Phase 2	4	L1, L2, L3
7	Result Phase II	2	L1, L2, L3, L4
8	Testing	2	L1, L2, L3, L4
9	Result validation	2	L1, L2, L3, L4, L5
10	Report Writing	4	L1, L2
	Total Hours	30	

Books and References:

Sr. No	Title	Authors	Publisher	Edition	Year
1	Professional Android 4 Application Development	RETO MEIER	Wrox publication	3rd	2012
2	Android Security attack and defenses, by CRC Press	Abhishek Dubey, Anmol Misra	CRC Press	1st	2013
3	Beginning Android Application Development	Wei-meng Lee	Wrox publication	1st	2011
4	Android Application Development For Dummies	Michael Burton, DonnFelker	John Wiley & Sons	2nd	2012
5	Android Cookbook	Ian F. Darwin	O'Reilly Media	1st	2011

Online Resources:

Sr. No.	Website Name	URL	Modules Covered
1.	tps://developer.android.com www.tutlane.com www.tutorialspoint.com	tps://developer.android.com/training/basics/firstapp tps://www.tutlane.com/tutorial/android/android-introduction tps://www.tutorialspoint.com/android/android_environment_setup.htm tps://www.tutorialspoint.com/android/android_application_components.h	M1
2.	www.tutlane.com www.tutorialspoint.com	tps://www.tutlane.com/tutorial/android/android-ui-controls-textview-itext-radio-button-checkbox tps://www.tutorialspoint.com/android/android_acitivities.htm	M2
3.	www.udemy.com www.coursera.org/ www.tutlane.com www.tutorialspoint.com	tps://www.udemy.com/learn-android-application-development-y/ tps://www.coursera.org/specializations/android-app-development tps://www.tutorialspoint.com/android/android_intents_filters.htm	M3
4.	www.tutlane.com www.tutorialspoint.com	tps://www.tutlane.com/tutorial/android/android-content-providers-with-examples tps://www.tutlane.com/tutorial/android/android-sqlite-database-with-examples tps://www.tutorialspoint.com/android/android_json_parser.htm	M4

5.	www.tutlane.com www.tutorialspoint.com	https://www.tutlane.com/tutorial/android/android-google-maps-api-with-examples https://www.tutorialspoint.com/android/android_location_based_services.html https://www.tutorialspoint.com/android/android_camera.html https://www.tutlane.com/tutorial/android/android-audio-media-player-with-examples	M5
6.	www.tutlane.com www.tutorialspoint.com	https://www.tutlane.com/tutorial/android/android-test-app-on-real-device-mobile-phone https://www.tutlane.com/tutorial/android/android-publish-app-on-google-play-store	M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic Student Development (CBCGS- HME 2023)
TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

BE Information Technology								B.E (SEM: VII)		
Course Name: User Interaction Design								Course Code: PEC-IT 7015		
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/Or al (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR	TW	150
3	-	2@	5	4	20	20	60	25	25	
ISE: In-Semester Examination - Paper Duration – 1 Hour										
IE: Innovative Examination										
ESE: End Semester Examination - Paper Duration - 2 Hours										
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: Database Management System, Data Warehouse and Mining, Machine Learning.										

Course Objective: To inculcate interactive design skills in students thereby stressing on importance of good interface design, understand the importance of human psychology as well as social and emotional aspect in designing good interfaces, learn the techniques of data gathering, establishing requirements, analysis, data interpretation and techniques for prototyping and evaluating user experiences and bring out the creativity in each student – build interactive applications that are usable, effective and efficient for intended users.

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	Recognize the value of interface design and distinguish between good and bad designs in order to improve high-quality system design.	L1, L2, L3	1,4,11,12	1,2,3
2	Identify fundamental aspect of interaction design to develop a conceptual model & know importance of understanding users & their cognitive aspects.	L1,L2,L3,L4	2,3,5,6,7,8,11	2,3
3	Illustrate Process of Interaction Design and analyze practical issues in it.	L1,L2,L3,L4	1,3,4,5,6,9,10,12	1,3
4	Apply standard data Gathering techniques for data collection & Illustrate Task Description & Task Analysis	L1,L2,L3,L4,L5,L6	2,5,8,9,10	2,3
5	Apply design concepts to develop prototype & conceptual design for developing interactive software product.	L1,L2,L3,L4,L5,L6	2,3,4,5,9,12	1,3
6	Understand and continue to apply the principles of design, the user-centric approach, and interaction design evaluation.	L1, L2, L3, L4	3,5,8,11,12	3

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction to Interaction Design	7	L1,L2,L3
	What is interaction design?, Why is this important?, Good and Poor Design, User, understanding users' need, , The Process Of Interaction Design, what do professionals do in the ID business?, Goals of Interaction Design , Usability, Usability Goals, User Experience goals.		
2	Understanding and Conceptualizing Interaction & Users	8	L1,L2,L3,L4
	Conceptualizing Interaction: Understanding the Problem Space and Conceptualizing Design, Conceptual Models: 2.3.1 Conceptual models based on activities, Conceptual models based on Objects, Interface metaphors, Interaction paradigms. Understanding Users: Cognition, Applying knowledge from physical word to Digital world, Conceptual Frameworks for Cognition: Mental Model, Information Processing, External Cognition.		
3	The process of interaction design	8	L1,L2,L3,,L4
	Introduction, Activities of Interaction design, Three key characteristics of the interaction design process Practical issues: Who are the users? What do we mean by “needs”? How do you generate alternative designs? How do you choose among alternative designs? Lifecycle Models: Life Cycle model of Interaction Design, Life Cycle model in Software Engineering, Life Cycle Model in HCI, How these models are interrelated?		
4	Data Gathering, Establishing Requirements & Task Analysis	7	L1,L2,L3,,L4,L5, L6
	What, How & why? What are requirements, Five Key Issues, Techniques for Data Gathering, guidelines of Data Gathering, Data Interpretation, Analysis and Presentation, Task Description: Scenario, Use cases & Essential use cases and Task Analysis		
5	Prototyping, Construction & Design	8	L1,L2,L3,L4,L5, L6
	Prototyping & Construction: What is Prototype?, Why Prototype?, Low-fidelity, High-fidelity , Construction from design to implementation, Conceptual Design: From requirements to first Design, perspectives for developing conceptual model, Scenarios & prototypes in Conceptual model. Physical Design: Guidelines , Different kinds of widgets		
6	User Centred Approach, Design rules and Evaluation	7	L1,L2,L3,L4
	User Centered approaches to Interaction Design: Why is it important to involve users, understanding users' work, users in design: Participatory Design, PICTIVE & CARD. Design principles: Principles to support Usability, Standards and Guidelines, Golden rules and Heuristics. Evaluation: What, Why & when to evaluate, case study on evaluation.		

Total Hours		45	

Capstone Project Guide Lines

1. The mini project work is to be conducted by a group of three students
2. Each group will be associated with a subject Incharge/ mini project mentor. The group should meet with the concerned faculty during Laboratory hours and the progress of work discussed must be documented.
3. The students may design an app that helps you use your screens less or Design a better interface for existing systems.
4. Students can:
 - I. Research the design causes of an increase in local crime,
 - II. Find a local charity and redesign a donation portal for them,
 - III. Write a case study about a time when you had to wait in line and how design could have sped up that process,
 - IV. Design a website where UX designers can post design patterns and test results etc...
5. Each group along with the concerned faculty shall identify a potential problem statement for application or web designs, on which the study and implementation is to be conducted.
7. Each group may present their work in various project competitions and paper presentations.
8. A detailed report is to be prepared as per guidelines given by the concerned faculty.

Capstone Project Hours Distribution:

Sr. No.	Work to be done	No. of Hours	Cognitive levels of attainment as per Bloom's Taxonomy
1	Study Research papers, articles, mini project title Identification	4	L1, L2
2	Project Title finalization and development of Modules	2	L1, L2
3	Design methodology and tools for implementation	4	L1, L2
4	Implementation of Modules phase 1	4	L1, L2, L3
5	Result Phase I	2	L1, L2, L3, L4
6	Implementation of Modules Phase 2	4	L1, L2, L3
7	Result Phase II	2	L1, L2, L3, L4
8	Testing	2	L1, L2, L3, L4
9	Result validation	2	L1, L2, L3, L4, L5
10	Report Writing	4	L1, L2
	Total Hours	30	

Books and References:

S. No.	Title	Authors	Publisher	Edition	Year
1.	Interaction Design	J. Preece, Y. Rogers and H. Sharp	John Wiley & Sons		2002
2.	Human Computer Interaction	Alan Dix, Janet Finlay, Gregory D Abowd, Russell Beale	Pearson	Third	2009
3.	About Face3: Essentials of Interaction design	Alan Cooper, Robert Reimann, David Cronin	Wiley publication.		2007
4.	An Introduction to GUI Design Principles and Techniques	Wilbert O. Galitz	Wiley Computer Publishing	Second	2002
5.	The Design of Everyday Things	Don Norman	Basic Books		2013

Online References:

Sr. No.	Website Name	URL	Modules Covered
1.	www.interaction-design.org	https://www.interaction-design.org/literature/topics/ui-design	M1
2.	www.interaction-design.org	https://www.interaction-design.org/literature/article/what-is-interaction-design	M2
3.	www.studocu.com	https://www.studocu.com/en/document/university-of-southampton/interaction-design/lecture-notes/lecture-notes-lectures-1-9-interaction-design/691675/view	M3
4.	ocw.mit.edu	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-831-user-interface-design-and-implementation-spring-2011/lecture-notes/	M4
5.	www.toptal.com	https://www.toptal.com/designers/interactive/interaction-design-principles	M5
6.	theblog.adobe.com	https://theblog.adobe.com/15-rules-every-ux-designer-know/	M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic Student Development (CBCGS- HME 2023)
TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

BE Information Technology								B.E (SEM: VII)			
Course Name: Soft Computing								Course Code: PEC-IT 7021			
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)						
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation						
Hours Per Week					Theory (100)			Practical/Oral (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ES E	OR	TW	150	
3	-	-	3	3	20	20	60	25	25		
ISE: In-Semester Examination - Paper Duration – 1 Hour											
IE: Innovative Examination											
ESE: End Semester Examination - Paper Duration - 2 Hours											
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).											
Prerequisite: Algorithm, Programming skills in C, C++, or Java, MATLAB, Python etc.											

Course Objective: To inculcate interdisciplinary engineering skills this course will cover fundamental concepts used in soft computing. The concepts of Fuzzy logic (FL) will be covered first, followed by Artificial Neural Networks (ANNs) and optimization techniques using Genetic Algorithm (GA). Applications of Soft Computing techniques to solve a number of real life problems will be covered to have hands on practices. In summary, this course will provide exposure to theory as well as practical systems and software used in soft computing.

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	CO1: List the facts and outline the different process carried out in fuzzy logic, ANN and Genetic Algorithms.	L1,L2	1, 2, 5, 11	1
2	CO2: Explain the concepts and meta-cognitive of soft computing.	L1,L2	1, 3, 5, 11	1, 2
3	CO3: Apply Soft computing techniques the solve character recognition, pattern Classification, regression and similar problems.	L1,L2,L3	1, 3, 5, 8	1, 2
4	CO4 : Outline facts to identify process/procedures to handle real world problems using soft computing.	L1,L2,L3,L4,L5,L6	1, 2, 3, 4, 5	1, 2

5	CO5: Evaluate various techniques ,features and operational properties of Genetic Algorithm.	L1,L2,L3,L4,L5,L6	1, 3, 5, 6	1, 2
6	CO6: Discuss characteristics, properties of hybrid system to revise the principles of soft computing in various applications.	L1,L2,L3,L4,L5,L6	1, 3, 5, 7, 10, 11	1, 2, 3

Detailed Syllabus:

Module No.	Topics	Hr s.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction	3	L1,L2
	What is Soft Computing? Difference between Hard and Soft computing, Requirement of Soft computing, Major Areas of Soft Computing, Applications of Soft Computing.		
2	Fuzzy Systems	8	L1,L2
	Fuzzy Set theory, Fuzzy versus Crisp set, Fuzzy Relation, Fuzzification, Minmax Composition, Defuzzification Method, Fuzzy Logic, Fuzzy Rule based systems, Predicate logic, Fuzzy Decision Making, Fuzzy Control Systems, Fuzzy Classification		
3	Neural Network I	10	L1,L2,L3
	Artificial Neural Networks: An Introduction, Supervised Learning: Introduction and how brain works, Neuron as a simple computing element, The perceptron, Backpropagation networks: architecture, multilayer perceptron, backpropagation learning-input layer, accelerated learning in multilayer perceptron, The Hopfield network, Bidirectional associative memories (BAM), RBF Neural Network.		
4	Neural Network II	10	L1,L2,L3,L4,L5,L6
	Unsupervised Learning: Hebbian Learning, Generalized Hebbian learning algorithm, Competitive learning, Self- Organizing Computational Maps: Kohonen Network. Building blocks of Adaptive Resonance, Substrate of resonance, Structural details of the resonance Model, Adaptive Resonance Theory I, Adaptive Resonance Theory II. Special Networks: Simulated Annealing Network, Boltzmann Machine, Gaussian Machine, Cauchy Machine.		
5	Genetic Algorithms	6	L1,L2,L3,L4,L5,L6
	Traditional Optimization and search techniques, Genetic algorithm and search space, History of Genetic Algorithms (GA), Working Principle, Various Encoding methods, Fitness function, GA Operators- Reproduction, Crossover, Mutation, Convergence of GA, Bit wise operation in GA, Nature inspired Algorithms : Particle Swarm Algorithm and Ant Colony Algorithm.		
	Hybrid Computing		

6	Introduction, Neuro-Fuzzy Hybrid Systems, Adaptive Neuro-Fuzzy Inference System (ANIFS): Introduction, ANFS Architecture, Hybrid Algorithm, ANFIS as a Universal Approximator, Simulation Examples: Two-input Sinc Function and Three Input Nonlinear Function Genetic Neuro-Hybrid Systems: Properties of Genetic Neuro-Hybrid Systems, genetic Algorithm based Back-propagation Network, Advantages of Neuro-Genetic Hybrids, Genetic Fuzzy Hybrid and Fuzzy Genetic Hybrid Systems Genetic Fuzzy Rule based Systems, Advantages of Genetic Fuzzy Hybrids.	8	L1,L2,L3,L4, L5,L6
Total Hours		45	

Capstone Project Guide Lines

1. The mini project work is to be conducted by a group of three students
2. Each group will be associated with a subject in charge/ mini project mentor. The group should meet with the concerned faculty during Laboratory hours and the progress of work discussed must be documented.
3. The students may do survey for different application which they can create project.
4. Each group along with the concerned faculty shall identify a potential problem statement, on which the study and implementation is to be conducted.
5. Each group may present their work in various project competitions and paper presentations.
6. A detailed report is to be prepared as per guidelines given by the concerned faculty.

Capstone Project Hours Distribution

Sr. No.	Work to be done	No. of Hours	Cognitive levels of attainment as per Bloom's Taxonomy
1	Study soft computing related Research papers, articles, mini project title Identification	4	L1, L2
2	Project Title finalization and development of Modules	2	L1, L2
3	Design methodology and tools for implementation	4	L1, L2
4	Implementation of Modules phase 1	4	L1, L2, L3
5	Result Phase I	2	L1, L2, L3, L4
6	Implementation of Modules Phase 2	4	L1, L2, L3
7	Result Phase II	2	L1, L2, L3, L4
8	Testing	2	L1, L2, L3, L4
9	Result validation	2	L1, L2, L3, L4, L5
10	Report Writing	4	L1, L2
	Total Hours	30	

Books and References:

Sr	Title	Authors	Publisher	Edition	Year
1	Principles of Soft Computing, Wiley India, 2007	S.N. Sivanandan and S.N. Deepa	Wiley India	Second	2007
2	Neuro-Fuzzy and Soft Computing, A Computational Approach to Learning and Machine Intelligence	J.-S. R. Jang, C. – T. Sun, E. Mizutani,	PHI Learning Private Limited	First	2014
3	Genetic Algorithms, in search, optimization and Machine Learning	David E. Goldberg	Pearson	First	1989

Online References:

.	Website Name	URL	Modules Covered
1	https://nptel.ac.in	https://nptel.ac.in/courses/106/105/106105173/	M1-M6
2	http://neuralnetworksanddeeplearning.com	http://neuralnetworksanddeeplearning.com/chap2.html	M3
3	https://www.analyticsvidhya.com/	https://www.analyticsvidhya.com/blog/2017/07/introduction-to-genetic-algorithm/	M5

B.E. Semester –VII
Choice Based Credit Grading Scheme with Holistic Student Development (CBCGS- HME2023)
TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

BE Information Technology								B.E (SEM : VII)			
Course Name: Blockchain Technologies								Course Code: PEC-IT 7022			
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)						
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation						
Hours Per Week					Theory (100)		Practical/Oral (25)		Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	IA	ESE	OR		TW	150	
3	-	2@	5	4	25	75	25		25		
IA: In-Semester Assessment- Paper Duration – 1.5 Hours ESE : End Semester Examination- Paper Duration - 3 Hours The weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely completion of practical (40%) and Attendance / Learning Attitude (20%) @ Capstone Project											
Prerequisite: Substantial programming experience, software engineering											

Course Objective: The course intends to deliver understanding of fundamentals of blockchain, list the concepts and blockchain technologies that can be used in application development

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy
1	Understand the components of blockchain and cryptocurrencies	L1,L2
2	Describe the concepts, technology and applications of Blockchain to be applied in the application	L1,L2
3	Distinguish and apply different consensus algorithm as per the requirement of application	L1,L2,L3
4	Get acquainted with different crypto currencies and their characteristics	L1,L2,L3,L4
5	Apply blockchain technology thinking to improve on existing products in IT	L1,L2,L3,L4,L5,L6
6	Design, build, and deploy a distributed application	L1,L2,L3,L4,L5,L6

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Basics of Blockchain	7	L1,L2
	Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance ,Introduction, Advantage over conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Blockchain application, Soft & Hard Fork, Private and Public blockchain.		
2	Distributed Ledger Technology	8	L1,L2
	Cryptography: Hash function, Digital Signature - ECDSA, Memory Hard Algorithm, Zero Knowledge Proof, Digital Trust, Asset, Transactions,Distributed Ledger Technology,Types of network Components of blockchain or DLT,Ledger Blocks,Blockchain,PKI and Cryptography,Private keys,Public keys, Hashing , Digital Signature		
3	Distributed Consensus	8	L1,L2,L3,L4
	Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy utilization and alternate		
4	Cryptocurrency	7	L1,L2,L3,L4
	Currency ,Double Spending, Cryptocurrenc, P2P Payment Gateway,Wallet, Mining,public blockchain and private blockchain,Other Crypt currencies		
5	Ethereum and DAPPs	7	L1,L2,L3,L4,L5,L6
	Ethereum network , EVM ,Transaction fee ,Mist , Ether, gas , Solidity, Smart contracts ,Truffle ,Web3, Design and issue Cryptocurrency , Mining , DApps , DAO		
	Hyperledger Fabric	8	L1,L2,L3,L4,L5,L6

6	Introduction to Hyperledger , What is Hyperledger , Why Hyperledger ,Where can Hyperledger be used , Hyperledger Architecture , Membership , Blockchain ,Transaction , Chaincode , Hyperledger Fabric ,Features of Hyperledger		
Total Hours		45	

Capstone Project Guide Lines

1. The mini project work is to be conducted by a group of three students
2. Each group will be associated with a subject Incharge/ mini project mentor. The group should meet with the concerned faculty during Laboratory hours and the progress of work discussed must be documented.
3. The students may do survey for different application which they can apply blockchain .
4. Students will do Installation, configuration and also try for Cross platform Integrated Development Environment
5. Students will try to Design and implement following points in their Mini Project
 - a) Front end
 - b) Middleware
 - c) Backend (Blockchain)
6. Each group along with the concerned faculty shall identify a potential problem statement for project development, on which the study and implementation is to be conducted.
7. Each group may present their work in various project competitions and paper presentations.
8. A detailed report is to be prepared as per guidelines given by the concerned faculty.

Capstone Project Hours Distribution:

Sr. No.	Work to be done	No. of Hours	Cognitive levels of attainment as per Bloom's Taxonomy
1	Study Research papers, articles, mini project title Identification	4	L1, L2
2	Project Title finalization and development of Modules	2	L1, L2
3	Design methodology and tools for implementation	4	L1, L2
4	Implementation of Modules phase 1	4	L1, L2, L3
5	Result Phase I	2	L1, L2, L3, L4
6	Implementation of Modules Phase 2	4	L1, L2, L3
7	Result Phase II	2	L1, L2, L3, L4
8	Testing	2	L1, L2, L3, L4

9	Result validation	2	L1, L2, L3, L4, L5
10	Report Writing	4	L1, L2
	Total Hours	30	

Books and References:

Sr. No	Title	Authors	Publisher	Edition	Year
1	Bitcoin and cryptocurrency technologies: a comprehensive introduction.	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder	Princeton University Press, 2016.	Second	2016
2	Mastering Bitcoin	Andreas Antonopoulos	O'REILLY	First	2014
3	Mastering Blockchain	Imran Bashir	Packt Publishing	First	2017

Online References:

Sr. No.	Website Name	URL	Modules Covered
1	https://www.coursera.org	https://www.coursera.org/learn/blockchain-professionals	M1,M2,M3,
2	https://www.coursera.org	https://www.coursera.org/learn/smarter-contracts	M1.M2,M3
3	https://www.coursera.org	https://www.coursera.org/learn/blockchain-foundations-and-use-cases	M1.M2,M3
4	https://www.dappuniversity.com	https://www.dappuniversity.com/articles/the-ultimate-ethereum-dapp-tutorial	M4,M5
5	https://www.hyperledger.org	https://www.hyperledger.org/use/fabric	M6
6	https://www.tutorialspoint.com	https://www.tutorialspoint.com/blockchain/index.htm	M4,M5,M6

7	https://www.guru99.com	https://www.guru99.com/blockchain-tutorial.html	M1,M2,M5,M6
8	https://www.javatpoint.com/	https://www.javatpoint.com/blockchain-tutorial	M1,M2,M5,M6
9	https://ghostvolt.com	https://ghostvolt.com/articles/blockchain_intro.html	M1,M2,M3,M4,M6
10	https://hackr.io/	https://hackr.io/blog/blockchain-programming-beginners-guide	M4,M5,M6

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic & Multidisciplinary Education (CBCGS-HME 2023)
Proposed TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

BE Information Technology					B.E (SEM: VII)					
Course Name: Web Analytics and Information Retrieval					Course Code: : PEC-IT 7023					
Teaching Scheme (Program Specific)					Examination Scheme (Academic)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/Oral / Presentation (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	I E	ES E	OR	TW	150
3	-		3	3	20	20	60	25	25	
ISE: In-Semester Examination - Paper Duration – 1 Hour										
IE: Innovative Examination										
ESE: End Semester Examination - Paper Duration - 2 Hours										
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: : Basic knowledge of programming, statistics, databases, web technologies, and data analytics.										

Course Objective: This course introduces students to the foundational concepts and applications of Web Analytics and Information Retrieval. It emphasizes the collection, analysis, and interpretation of web data, SEO strategies, and the implementation of information retrieval models to enhance digital presence and user experience.

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	Explain the fundamental concepts of web analytics and information retrieval systems.	L1, L2	1, 2	1
2	Apply appropriate web analytics tools and interpret collected data effectively.	L1, L2, L3	1,2, 3, 5, 11	1, 2
3	Analyze user behaviour and SEO performance using key metrics and techniques.	L1, L2, L3, L4	2, 3, 5	1, 2

4	Evaluate different information retrieval models and implement ranking algorithms.	L1, L2, L3, L4	1, 2, 3, 5	1, 2
5	Design and execute optimization strategies using A/B testing, personalization, and recommendation systems.	L1, L2, L3, L4, L5	3, 4, 5, 11	2, 3
6	Develop real-time analytics dashboards using big data frameworks while addressing ethical considerations.	L1,L2,L3,L4,L5,L6	3, 5, 6, 10	2, 3

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction to Web Analytics and Information Retrieval	6	L1,L2
	Overview of Web Analytics: metrics, KPIs, user behaviour, Basics of Information Retrieval: indexing, querying, ranking, Web data sources: server logs, APIs, clickstreams, Components of IR systems: document representation, retrieval models, Applications in IT: search engines, digital marketing, user behaviour analysis		
2	Web Analytics Techniques and Tools	8	L1,L2,L3
	Web metrics: page views, bounce rate, conversion rate, dwell time, Data collection: server logs, cookies, JavaScript tags, Tools: Google Analytics, Adobe Analytics, Matomo (overview), Data visualization and dashboard reporting, Applications: user tracking, campaign analysis, performance monitoring		
3	SEO and User Behaviour Analysis	7	L1,L2,L3,L4
	SEO fundamentals: on-page, off-page, and technical SEO, SEO metrics: organic traffic, bounce rate, backlinks, domain authority, Keyword research and content optimization, Tools: Google Search Console, Ahrefs, SEMrush, User segmentation: demographic, geographic, behavioural, Funnel analysis and user journey mapping Applications: website visibility, engagement optimization		
4	Information Retrieval Models and Ranking Algorithms	8	L1,L2,L3,L4
	IR models: Boolean, Vector Space, Probabilistic, Ranking algorithms: TF, IDF, TF-IDF, BM25, PageRank, HITS (Hyperlink-Induced Topic Search) Query processing: query formulation, expansion, Evaluation metrics: precision, recall, F1-score, MAP, Applications: e-commerce search, enterprise search		

5	Advanced Analytics and Recommender Systems	8	L2,L3,L4,L5
	A/B testing and multivariate testing, Conversion Rate Optimization (CRO) Recommender systems: collaborative filtering, content-based filtering, Predictive analytics: intro to clustering and classification, Applications: personalization, marketing ROI, churn prediction		
6	Emerging Trends in Web and Information Analytics	8	L1,L2,L3,L4,L5
	Real-time analytics and stream processing, Semantic search and embeddings: Word2Vec, BERT, Social media and mobile analytics, Big data frameworks: Hadoop, Spark, Ethical issues: data privacy (GDPR, CCPA), consent management Applications: smart cities, IoT analytics, voice search		
Total Hours		45	

Capstone Project Guide Lines

1. The mini project work is to be conducted by a group of three students
2. Each group will be associated with a subject Incharge/ mini project mentor. The group should meet with the concerned faculty during Laboratory hours and the progress of work discussed must be documented.
3. Students may conduct a survey of different applications where Web Analytics and Information Retrieval techniques can be applied (e.g., e-commerce, blogging platforms, CRM, content recommendation systems).
4. Students will perform installation, configuration, and usage of latest Web Analytics and SEO tools (e.g., Google Analytics, Ahrefs, SEMrush, Google Search Console, Google Data Studio).
5. Students will try to consider following points in their Mini Project
 - a) Web data collection methods
 - b) Metric analysis (traffic, bounce rate, conversions)
 - c) Visualization of user behaviour and funnels
 - d) SEO audit and performance tracking
 - e) Keyword ranking and optimization strategy
6. Each group along with the concerned faculty shall identify a potential problem statement for project development, on which the study and implementation is to be conducted.
7. Each group may present their work in various project competitions and paper presentations.
8. A detailed report is to be prepared as per guidelines given by the concerned faculty.

Capstone Project Hours Distribution:

Sr. No.	Work to be done	No. of Hours	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Study WAIR tools, research papers, articles; identify mini project topics	4	L1, L2

2	Project title finalization and development of modules	2	L1, L2
3	Design methodology and choose tools (e.g., Google Analytics, Ahrefs)	4	L1, L2
4	Implementation of modules – Phase 1 (data collection, tracking setup)	4	L1, L2, L3
5	Result Phase I	2	L1, L2, L3, L4
6	Implementation of modules – Phase 2 (SEO audit, behaviour analysis)	4	L1, L2, L3
7	Result Phase II	2	L1, L2, L3, L4
8	Testing and metric validation (bounce rate, CTR, session analysis)	2	L1, L2, L3, L4
9	Result validation	2	L1, L2, L3, L4, L5
10	Report Writing	4	L1, L2
	Total Hours	30	

Books and References:

Sr. No.	Title	Authors	Publisher	Edition	Year
1	Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity	Avinash Kaushik	Sybex (Wiley)	1st Edition	2009
2	Digital Marketing Analytics: Making Sense of Consumer Data in a Digital World	Chuck Hemann, Ken Burbary	Que Publishing	2nd Edition	2018
3	Introduction to Information Retrieval	Christopher D. Manning, Prabhakar Raghavan, Hinrich Schütze	Cambridge University Press	1st Edition	2008
4	Search Engines: Information Retrieval in Practice	Bruce Croft, Donald Metzler, Trevor Strohman	Pearson	1st Edition	2010
5	SEO 2023: Learn Search Engine Optimization	Adam Clarke	Kindle/Independent	Latest Edition	2023
6	Google Analytics Breakthrough: From Zero to Business Impact	Feras Alhlou, Shiraz Asif, Eric Fettman	Wiley	1st Edition	2016
7	Practical Web Analytics for User Experience	Michael Beasley	Morgan Kaufmann	1st Edition	2013

Online References:

Sr. No.	Website Name	URL	Modules Covered
1	Google Analytics Academy	https://analytics.google.com/analytics/academy	M2, M3, M4
2	Ahrefs Blog & Academy	https://ahrefs.com/blog	M5, M6
3	SEMrush SEO Learning	https://www.semrush.com/academy/	M5
4	Google Search Central	https://developers.google.com/search	M5
5	Google Data Studio	https://lookerstudio.google.com	M4
6	Tableau Public	https://public.tableau.com	M4
7	Moz Beginner's Guide to SEO	https://moz.com/beginners-guide-to-seo	M5
8	Towards Data Science (Web Analytics)	https://towardsdatascience.com/tagged/web-analytics	M1, M3, M6
9	W3Schools Analytics & SEO	https://www.w3schools.com	M1, M2, M3, M5
10	HubSpot Marketing and Analytics Blog	https://blog.hubspot.com/marketing	M1, M2, M5, M6

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic Student Development (CBCGS- HME 2023)
TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

BE Information Technology					B.E (SEM: VII)					
Course Name : Management Information System					Course Code: PEC- IT 7024					
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR	TW	
3	-	-	3	3	20	20	60	25	25	150
ISE: In-Semester Examination - Paper Duration – 1 Hour										
IE: Innovative Examination										
ESE: End Semester Examination - Paper Duration - 2 Hours										
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: : Database Design and Management										

Course Objective: The course intends to deliver the role of Management in Information Systems to understand the impact of these systems within an Organization to improve business performance and decision making. It analyzes typical functional information systems, principal tools and technologies for accessing information from databases & interpreting Ethical issues & Privacy for the same.

Course Outcomes : Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	Explain how information systems Transform Business	L1, L2	1, 6, 11	1
2	Understand about Data and Knowledge Management	L1, L2, L3	1, 2, 5, 11	1
3	Analyze the Ethical issues and Privacy in Information Systems	L1, L2, L3, L4	1, 2, 4, 7	1
4	Understand the principal tools and technologies for accessing information from databases to improve business performance and decision making	L1, L2, L3	1, 2, 5	1
5	Analyze the types of systems used for enterprise-wide knowledge management and how they provide value for businesses	L1, L2, L3, L4	1, 2, 3, 5	1,3
6	Analyze the impact of information systems have on an organization	L1, L2, L3, L4	1, 2, 6, 11	1,3

Detailed Syllabus:

Module No.	Topics	Hours	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction To Information Systems (IS)	6	L1, L2
	Computer Based Information Systems, Impact of IT on organizations, Importance of IS to Society. Organizational Strategy, Competitive Advantages and IS Management Information System: Case Study of Amazon.Com		
2	Data and Knowledge Management	9	L1, L2, L3
	Database Approach, Big Data, Data warehouse and Data Marts, Knowledge Management Business intelligence (BI): Managers and Decision Making, BI for Data analysis and Presenting Results Case Study: Netflix uses big data to suggest shows based on what people watch and like, helping to keep users engaged.		
3	Ethical issues and Privacy	8	L1, L2, L3, L4
	Information Security- Goals of IS: Confidentiality, Integrity, Availability (CIA Triad), Security policies and governance in organizations, Threats to Information Systems , Types of Threats to Information Systems and Security Controls		
4	Social Computing (SC)	7	L1, L2, L3
	Web 2.0 and 3.0, SC in business-shopping, Marketing, Operational and Analytic CRM, E-business and E-commerce – B2B B2C. Mobile commerce.		
5	Wired and Wireless Technology	7	L1, L2, L3, L4
	Computer Networks Wired and Wireless Technology, Pervasive computing, Cloud computing model. Network security in wireless environments, Virtual Private Networks (VPNs) and firewalls		
6	Information System within Organization	8	L1, L2, L3, L4
	Transaction Processing Systems, Functional Area Information System, ERP and ERP support of Business Process. Acquiring Information Systems and Applications: Various System development life cycle models Case Study: IBM used a prototyping method to develop banking software by first creating a small working model.		
	Total Hours	45	

Books and References:

Sr. No.	Title	Authors	Publisher	Edition	Year
1.	Management Information Systems	Kelly Rainer, Brad Prince	Wiley	Sixth Edition	2011
2.	Management Information Systems	K.C. Laudon and J.P. Laudon	Prentice Hall	Tenth Edition	2007
3.	Managing Information Systems: Strategy and Organization	D. Boddy, A. Boonstra	Prentice Hall	Tenth Edition	2008

Online References:

Sr. No.	Website Name	URL	Modules Covered
1	https://www.tutorialspoint.com/index.htm	https://www.tutorialspoint.com/management_information_system/	M1
2	https://www.tutorialspoint.com/index.htm	https://www.tutorialspoint.com/dwh/dwh_data_warehousing.htm	M2
3	https://www.tutorialspoint.com/index.htm	https://www.tutorialspoint.com/management_information_system/mis_security_and_ethical_issues.htm	M3
4	https://www.aller.in.com	https://www.aller.in.com/blog/how-web-3-0-will-be-different-from-web-2-0-for-businesses	M4
5	https://pressbooks.com/	https://bus206.pressbooks.com/chapter/chapter-13-future-trends-in-information-systems/	M5
6	https://www.tutorialspoint.com	https://www.tutorialspoint.com/what-is-the-information-system	M6

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic Student Development (CBCGS- HME 2023) TCET
Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. (Information Technology)							B.E. (SEM: VII)			
Course Name: Internet of Everything							Course Code: PEC-IT-7025			
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)		Practical/Oral (50)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	IA	ESE	OR	TW	150	
3	-	2 @	5	4	25	75	25	25		
IA: In-Semester Assessment- Paper Duration – 1.5 Hours ESE: End Semester Examination- Paper Duration - 3 Hours Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%). @: Professional Elective Courses Lab will be conducted in the form Capstone Project										
Prerequisite: IOT Lab, Sensor Lab, Wireless Network										

Course Objective: The course intends to deliver the fundamentals of IOT concepts used for smart city development, IoT technologies, applications, protocols, and analytics of data in IOT. Also make real life application with the help of different design platforms used for an embedded systems application like Arduino and raspberry-pi

Course Outcomes: Upon completion of the course student will be able to:

Sr.No	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy
1	Apply the concepts of MPMC & IOT	L1, L2,L3
2	Identify & Understand the IOE Enabling technology & RFID Technology	L1, L2,L3

3	Apply Knowledge on RFID Case Study & different applications	L1, L2,L3
4	Analysis and evaluate protocols used in IOT	L1, L2,L3,L4
5	Design and develop smart city in IOT	L1, L2,L3,L4, L5,L6
6	Analysis and evaluate the data received through sensors in IOT	L1, L2,L3,L4,L5

Detailed Syllabus:

Module NO.	Topics		Cognitive levels of attainment as per Bloom's Taxonomy
	Prerequisite		
	Introduction to Arduino, Raspberry Pi. Basics of Arduino programming, Extended Arduino libraries. Sensors, Sensor Types and Interfacing: Temperature, Pressure, Humidity sensors & Sensor family.		L1,L2
	Introduction		
	Introduction to IOT, History of IOT, Objects/things in the IOT, Identifier in IOT, Enabling Technologies in IOT: Identification Technology, Sensing and actuating, Other technologies, Connected object's communication, about the Internet in IOT.		L1, L2,L3
	RFID Technology		
	Introduction to RFID, IOT & RFID Technology, Components of RFID System: RFID Tag, Types of Tags, RFID Readers. Principal of RFID, RFID middleware, Issues.		L1,L2,L3
	RFID Application and Challenges		

	RFID Applications: logistics and supply chain, production, monitoring and maintenance, product safety, quality and information, access control and tracking and tracing of individuals, payment, loyalty, household etc., Challenges of RFID Technology , Hardware, Hardware issues, protocols: pure aloha, slotted aloha, frame slotted aloha, tree protocols, tree splitting algorithms, binary search algorithms, bitwise arbitration protocols. Main query tree protocols.		L1, L2, L3,L4
	Wireless Sensor Networks Introduction to wireless sensor Network, Components of WSN, Types of Wireless Sensor Network Node: Connecting Nodes, Networking Nodes. Architecture of WSN, Securing Communication, Standards and Fora. Networking and Internet - IP Addressing, Protocols - MQTT, CoAP, REST Transferring data. WSN Protocols.		L1, L2, L3, L4,L5
	Mobility and Settings Introduction, localization, mobility management, localization and handover management, technology considerations, performance evaluation, simulation setup, performance results. Identification of IOT (Data formats, IPv6, identifiers and locaters, tag etc.)		L1, L2, L3, L4, L5
	Tools & Application for IOE Introduction, Apache, Hadoop, Using Hadoop MapReduce for Batch Data Analysis, Using Apache Storm for Real-time Data Analysis, Agriculture & Health Monitoring Case Study. Tools for IOT: Chef, Chef Case studies, Puppet, Puppet Case Study		L1, L2, L3, L4, L5
	Total Hrs		

Capstone Project:

Work to be done	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
Survey for different IoT application using Raspberry pi/Arduino etc	2	L1,L2,L3
IoE Project Title Identification	2	L1,L2,L3
Identify the Hardware and software requirement for their mini project problem statement.	4	L1,L2,L3
Prototype/Design your own circuit board using Raspberry pi/Arduino/sensors/Rfid	4	L1,L2,L3
Working Implementation and do coding to for input devices on board.	6	L1,L2,L3,L4
Create and interface using Web to publish or remotely access the data on Internet.	4	L1,L2,L3,L4
Present work in various project competitions / paper presentations etc	4	L1,L2,L3,L4,L5
Testing of Mini Project/case Study	2	L1,L2,L3,L4,L5
Reports Design & Preparation	2	L1,L2,L3,L4,L5
Total Hours	30	

Books and References:

Sr. No.	Title	Authors	Publisher	Edition	Year
1.	Internet of Things connecting objects to the web	Hakima Chaouchi	Wiley	1st	2010
2.	Internet of Things (A Hands-on Approach)	Arshdeep Bhaga and Vijay Madiseti.	--	---	2014
3.	The Internet of Things	Samuel Greengard	MIT Press	2nd	2015

4.	RFID and the Internet of Things	Herve chabanne	Wiley	1st	2013
5.	Fundamentals of Sensor Network Programming: Applications and Technology	S. Sitharama Iyengar, Nandan Parameshwaran, Vir V. Phoha, N. Balakrishnan, Chuka D. Okoye	John Wiley & Sons	1st	2010
6.	Building the internet of things with ipv6 and mipv6, The Evolving World of M2M Communications	Daniel Minoli	John Wiley & Sons	1st	2013
7.	6LoWPAN: The Wireless Embedded Internet	Zach Shelby, Carsten Bormann	Wiley	1st	2009
8.	Interconnecting Smart Objects with IP: The Next Internet	Jean-Philippe Vasseur, Adam Dunkels, Morgan Kuffmann	Elsevier	1st	2010
9.	Designing the Internet of Things	Adrian McEwen (Author), Hakim Cassimally	John Wiley & Sons	1st	2013
10.	Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems	Dr. Ovidiu Vermesan, Dr. Peter Friess	River Publishers	1st	2013
11.	Internet of Things (A Hands-on-Approach)	Vijay Madiseti, Arshdeep Bahga	John Wiley & Sons	1st	2014
12	ARM System on chip Architecture	Steve Furber	Pearson	2nd	2012

13	Getting Started with Arduino: The Open Source Electronics Prototyping Platform (Make)	Massimo Banzi	O'Reilly Media	-	-
14	RaspberryPi Cookbook	Simon Monk	O'Reilly Media	-	-

Online Recourses:

Sr. No.	Website Name	URL	Modules covered
1.	https://www.nptel.ac.in	https://nptel.ac.in/courses/106105166/	M1,M2,M3, M4,M5,M6
2.	https://www.tutorialspoint.com	https://www.tutorialspoint.com/internet_of_things/internet_of_things_pdf	M1,M2,M3, M6
3	http://www.infocobuild.com	http://www.infocobuild.com/education/audio-video-courses/electronics/MicroprocessorsMicrocontrollers-IIT-Kharagpur/lecture-34.html	M5, M6
4.	http://www.listingtec.com	http://www.listingtec.com/nptel-iot-assignment-8-answers-introduction-to-internet-of-things/	M1,M4,M5, M6

B.E. Semester – VII
 Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)
 TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Product Design &Development								Course Code: OEC-7011		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: Data Structure, Software Engineering, Soft Skills and Interpersonal Communication										

Course Objectives:

1. Understand the Product Development Process:

- To introduce students to the stages and steps involved in product design and development.
- To provide a comprehensive overview of the product development lifecycle.

2. Enhance Design Thinking:

- To cultivate design thinking skills, encouraging creative problem-solving and innovative idea generation.
- To develop the ability to think critically and apply design principles effectively.

3. Master Prototyping Techniques:

- To equip students with the knowledge and skills to create prototypes using various rapid prototyping methods.
- To enable students to communicate and visualize design concepts effectively.

4. Materials and Manufacturing Knowledge:

- To educate students about materials selection criteria and their impact on product design.
- To introduce common manufacturing processes and design for manufacturability (DFM) principles.

5. Design Optimization and Evaluation:

- To teach techniques for optimizing designs based on cost, performance, and other relevant factors.

- To provide students with tools for simulation and analysis of product designs.

6. Product Lifecycle Management (PLM):

- To introduce the concept of Product Lifecycle Management (PLM) and its significance in modern engineering practices.
- To familiarize students with PLM software and collaborative product development.

Sr.No	Course Outcomes	RBT Level
1	Understand the fundamentals of mobile communications Analyse and Execute the Product Development Process: • Understand the stages of product development and execute them effectively. • Apply systematic approaches to problem-solving and decision-making in product design.	L1, L2, L3
2	Apply Design Thinking Principles: • Demonstrate creative thinking and innovation in generating and evaluating design concepts. • Incorporate user-centric design principles into product development.	L1, L2, L3
3	Create Prototypes and Visualizations: • Develop functional prototypes using rapid prototyping techniques. • Present and communicate design concepts through effective visualizations.	L1, L2, L3, L4
4	Make Informed Material and Manufacturing Choices: • Select materials based on design requirements and constraints. • Identify suitable manufacturing processes and consider DFM principles in design.	L1, L2, L3
5	Optimize Designs and Perform Evaluation: • Apply optimization techniques to improve product designs in terms of cost, performance, and sustainability. • Use simulation and analysis tools to evaluate and refine designs.	L1, L2, L3, L4
6	Utilize Product Lifecycle Management: • Understand the role of PLM in product development. • Use PLM software for data management and collaboration in the product design process.	L1, L2, L3, L4

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction to Product Design and Development	06	L1, L2, L3
	Introduction to Product Design and Development Product development process overview Importance of design thinking		
2	Concept Generation and Ideation	08	L1, L2, L3
	Brainstorming and idea generation methods		
	Concept selection and evaluation Feasibility analysis		
3	Prototyping and Product Visualization	09	L1, L2, L3, L4
	Types of prototypes and their importance Rapid prototyping techniques Product visualization and presentation		
4	Materials and Manufacturing Processes	08	L1, L2, L3
	Materials selection criteria Common manufacturing processes		
5	Design Optimization and Evaluation	07	L1, L2, L3, L4
	Design optimization techniques Simulation and analysis tools for design Ethical considerations in product design		
6	Product Lifecycle Management (PLM)	07	L1, L2, L3, L4
	Introduction to Product Lifecycle Management (PLM) Role of PLM in product design and development PLM software and data management Collaborative product development		
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Product Design and Development	Karl T. Ulrich and Steven D. Eppinger	McGraw-Hill Education.	-	-

2	The Art of Innovation: Lessons in Creativity from IDEO, America's Leading Design Firm	Tom Kelley and Jonathan Littman	Doubleday	1	2001
3	Materials Selection in Mechanical Design	Michael F. Ashby	McGraw-Hill Education	1	1994
4	Design for Manufacturability Handbook	James G. Bralla	McGraw-Hill Professional	-	-

B.E. Semester – VII
 Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)
 TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Alternative Fuels								Course Code: OEC-7012		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination										
ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: N/A										

Course Objectives: Course intend to provide the knowledge on various alternate fuels for IC engines.

Sr.No.	Course Outcomes	RBT Level
1	To analyse the impact of fossil fuels on environment	L1, L2
2	To distinguish between types of alternative fuels.	L1
3	To determine the quality of liquid biofuels.	L1, L2, L3, L4
4	To determine the quality of gaseous biofuels.	L1, L2, L3, L4
5	To analyse the impact of alternative fuels on environment	L1, L2, L3, L4
6	To demonstrate basics of hydrogen fuel	L1, L2

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Fossil Fuels to Alternative Fuels	08	L1, L2
	Reserves of Fossil fuels in India and globe, Disadvantages of Fossil fuels, Need of Alternative fuels, Types, Advantages, Sources of Alternative fuels..		
2	Advanced Liquid Biofuels	08	L1
	Raw material for biofuel production, Biomass and Waste, First and next generation biofuels, Conversion of lignocellulosic, algal biomass, and waste into biofuels and		
	chemicals, Production of Biodiesel, Bio alcohol, Jatropha Fuel Applications.		
3	Advanced Gaseous Biofuels	08	L1, L2, L3, L4
	Bio-CNG from sugarcane, Synthetic gas SynGas, generation of SynGas through plasma gasification of plastic waste, Applications.		
4	Hydrogen Technology	07	L1, L2, L3, L4
	Hydrogen as Alternative fuel, Hydrogen storage, hydrogen liquefaction, ortho and para hydrogen, Non-fossil Natural gas and methane, Applications.		
5	Fuel Cells	07	L1, L2, L3, L4
	Principle & operation of Fuel cells, Thermodynamics of fuel cells, types of fuel cells, comparison of fuel cell technologies, stack configurations and fuel cell systems, Applications		
6	Alternative Fuels and Environmental Impact	07	L1, L2
	Climate change, Benefits of alternative fuel to environment, Environmental impact assessment		
Total Hours		45	

Books and References:

1. Sunggyu Lee, James G. Speight, Sudarshan K. Loyalka, Handbook of Alternative Fuel Technologies, CRC Press
2. Vishwanathan B and M Aulice Scibioh, Fuel Cells, Universities Press, Hyderabad, 2006
3. Gupta R B, Hydrogen fuel: Production, transport and storage, Boca Raton : CRC Press, 2008
4. Fuel Cell System, edited by Leo J.M.J. Blomen and Michael N. Mugerwa, New York, Plenum Press, 1993.
5. Michael Frank Hordeski, Alternative Fuels-The Future of Hydrogen, The Fairmont Press, Inc., CRC Press
6. G.N. Tiwari, M.K. Ghosal, Fundamentals of Renewable Energy Sources, Alpha Science Intl. Ltd., 2007
7. Prabir Basu; Biomass gasification and pyrolysis: Practical design and theory; Elsevier, 2010
8. Sergio Capareda, Introduction to Biomass Energy Conversions, CRC Press, 2013
9. H S Mukunda, Understanding Clean Energy and Fuels from Biomass, Wiley India
10. Nijaguna, B.T., Biogas Technology, New Age International publishers (P) Ltd

Online References:

S. No.	Website Name	URL	Modules Covered
1	www.nptel.ac.in	https://nptel.ac.in	M1-M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)					B.E. / B.Tech. SEM:VII					
Course Name: Food Safety & Management					Course Code: OEC-7013					
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)		Practical/ Oral (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	
3	-	-	3	3	20	20	60	-	25	125
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours The weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely completion of Tutorial (40%) and Attendance/Learning Attitude (20%)										
Prerequisite: Database Design and Management										

Course Objectives: To acquire knowledge about quality and safety aspects of food. To learn about the various ways of evaluating and controlling food quality.

SN	Course Outcomes	RBT Level
1	Understanding the Scope for Food safety and factors affecting Food	L1, L2
2	Understand & analyse about Different Fungal Food fermentation	L1, L2, L3, L4
3	Apply & Analyse the Objective evaluation for Storage of Food and Fermentation in variable Food	L1, L2, L3, L4
4	Examine Food Additives & Food Adulteration Method of Food Preservation also conclude the methods	L1, L2, L3, L4
5	Assess Extrinsic factors and their impact over Food.	L1, L2, L3, L4, L5
6	Evaluate Food Preservation and its Significance	L1, L2, L3, L4, L5

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Food Quality Food Quality Meaning and definition of food quality Quality factors in foods, indicators of food quality. Meaning, importance and ways of Food Quality Assessment.	08	L1, L2
2	Sensory evaluation Physiological bases, sensory characteristics of foods, types, selection and training of sensory panel, requirements for sensory evaluation tests, types of tests, analysis and interpretation of sensory evaluation tests	07	L1, L2, L3, L4
3	Objective evaluation Basic guidelines, physical methods to evaluate volume, specific gravity, moisture, texture rheological characteristics, chemical analysis methods, microscopic methods, indices of microbial quality	07	L1, L2, L3, L4
4	Food Additives & Food Adulteration Brief overview, classification, guidelines for use, MAQ of food additives, toxicological studies, tests to determine safe level – Acute test, prolonged test, chronic test. Meaning, detection of common adulterants. PFA laws related to food adulteration	07	L1, L2, L3, L4
5	Extrinsic Factor Extrinsic Factors Affecting Microbial Growth and Survival in Food, Microbiological Examination of Food, Advances in Isolation and Enumeration of Microorganisms in Food	08	L1, L2, L3, L4, L5
6	Principles of Food Preservation and Significance Principles of Food Preservation and Significance, Preservation of Food by Physical Methods – Low and High Temperatures, Preservation of Food by Physical Methods – Radiations	08	L1, L2, L3, L4, L5
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Food Safety Management	Yasmine Motarjemi , Huub Lelieveld Dr.	Academic Press	First Edition	2014
2	Food Safety and Quality Management	Ronald F. Cichy, Ph.D. and JaeMin Cha, Ph.D.	Softbound,ISBN	Third Edition	2019.
3	Food Safety and Human Health	Robert W. Kapp, Jr, PhD	Academic Press	First Edition	2020

Online References:

S N	Website Name	URL	Modules
1	https://doi.org/10.1002/9781119053569.refs	https://www.foodsafetymagazine.com/magazine-archive1/junejuly-2014/why-dont-we-learn-more-from-our-mistakes/	M1
2	https://doi.org/10.1002/9781119053569.refs	http://www.foodqualityandsafety.com/article/the-cumin-scandal-accidental-or-fraudulent/..	M2
3	https://doi.org/10.1002/9781119053569.refs	http://www.ars.usda.gov/services/software/software.htm	M3
4	https://doi.org/10.1002/9781119053569.refs	https://www.food.gov.uk/sites/default/files/food-and-you-w4-combined-report.pdf..	M4
5	https://doi.org/10.1002/9781119053569.refs	https://www.foodsafetymagazine.com/magazine-archive1/augustseptember-2006/4-steps-to-strategic-sanitation-risk-assessment/	M5
6	https://doi.org/10.1002/9781119053569.refs	http://www.inspection.gc.ca/food/safe-food-production-systems/food-safety-enhancement-program/program-manual/eng/1345821469459/1345821716482?chap=4#s6c4	M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Design Thinking								Course Code: OEC-7014		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours The weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely completion of practical (40%) and Attendance / Learning Attitude (20%)										
Prerequisite: NIL										

Course Objectives: To inculcate interdisciplinary engineering skills in students for taking real time engineering problem available in our society/industry and to come-up with the grass root innovation, can be helpful to all level of human beings.

SN	Course Outcomes	RBT Level
1	Understand the importance of Design Thinking and Apply design thinking for product development	L1, L2
2	Evaluate through case studies and exercises to critically analyze the effectiveness of Design Thinking	L1, L2, L3, L4
3	Identify the needs for design of sustainable society	L1, L2, L3, L4
4	Apply standard problem-solving heuristics to aid in problem solving.	L1, L2, L3, L4, L5, L6
5	Apply design thinking tools to improve product design	L1, L2, L3, L4, L5, L6
6	Formulate and successfully communicate the solutions to problems.	L1, L2, L3, L4, L5, L6

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Design Thinking Introduction	08	L1, L2
	Introduction to Design Thinking, Need of Design Thinking, Traditional Problem Solving versus Design Thinking, phases of Design Thinking, Tools for Design Thinking, History of design thinking, 5 stages of Design Thinking Process, case study		
2	Design Thinking Exercise	08	L1, L2, L3, L4
	Formation of Team and aspects for the selection, Design activities through business Canvas model, brainstorming for the problem, good design and bad design, Multiple users, understanding user experience, design thinking in HR, MVP or prototyping customer reactions to prototype, Appeal for proposals, case study		
3	Design and Society, sustainability	08	L1, L2, L3, L4
	Community and collaboration, understanding contexts, needs and assess, meeting needs, functional context and consequence's, people centered design methodology, reconsider and arriving at the 'right' problem to solve, personas, Insight statements how might we questions Design and sustainability Sustainability practices for daily life ,perspective of engineering		
4	Design and industry	07	L1, L2, L3, L4, L5, L6
	Generating ideas, top five ideas, Bundle ideas and create concept, Understanding user needs, new materials used for manufacturing, rapid prototyping, mass production Design and collaboration Principles of collaboration, Team Partners, collaborating with unlikely partners ,feedback		
5	Innovation in Design	09	L1, L2, L3, L4, L5, L6
	Seven concerns of innovation, pitfalls of innovation, user feedback and development, positioning the product ,innovative templates, innovative domains		
6	Testing, Refining and Pitching		

	Importance of testing with people, conducting usability test, records, results, retest and redefine design, create pitch for design	05	L1, L4, L5, L6
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Strategies for Creative Problem Solving	H. S. Fogler and S. E. LeBlanc	Pearson,	Second	2008
2	Problem Solving & Comprehension	A. Whimbey and J. Lochhead	Lawrence Erlbaum, Mahwah,	Sixth	1999
3	The Design of Business: Why Design Thinking is the Next Competitive Advantage	Roger Martin	Harvard Business Press	First	2009
4	Design Thinking: Understand – Improve – Apply	Hasso Plattner, Christoph	Springer, 2011 (Unit III)	First	2011
5	Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School. (Unit IV).	Idris Mootee	John Wiley & Sons 2013	First	2013
6	Effective Problem Solving	M. Levine	Prentice Hall	Second	1994

Online References:

S. No.	Website Name	URL	Modules Covered
1	https://www.coursera.org	https://www.coursera.org/learn/uva-darden-design-thinkinginnovation	M1, M2, M3
2	http://www.cs.odu.edu	http://www.cs.odu.edu/~cs381/cs381content/problem_solving/problem_solving.html	M4, M5, M6

3	https://www.cs.vt.edu	https://www.cs.vt.edu/undergraduate/courses/CS2104	M1, M2, M3, M4, M5, M6
4	https://ryanstutorials.net	https://ryanstutorials.net/problem-solving-skills/	M3, M4
5	https://dschool.stanford.edu	https://dschool.stanford.edu/.../designresources/.../ModeGuideBOOTCAMP2010L.pdf	M1, M2, M3, M5
6	https://dschool.stanford.edu	https://dschool.stanford.edu/use-our-methods/	M4, M5, M6
7	https://www.interaction-design.org	https://www.interaction-design.org/literature/article/5-stages-in-the-design-thinking-process	M1, M2, M5, M6
8	http://www.creativityatwork.com	http://www.creativityatwork.com/design-thinking-strategy-for-innovation/49	M1, M2, M5, M6
9	https://www.nngroup.com	https://www.nngroup.com/articles/design-thinking/	M1, M2, M3, M4, M6
10	www.designthinkingformobility.org	www.designthinkingformobility.org/wp-content/.../10/NapkinPitch_Worksheet.pdf	M4, M5, M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)					B.E. / B.Tech. SEM:VII					
Course Name: Research Methodology					Course Code: OEC-7015					
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)		Practical/ Oral (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	
3	-	-	3	3	20	20	60	-	25	125
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: Discrete Mathematics										

Course Objectives:

1. To understand Research and Research Process
2. To acquaint students with identifying problems for research and develop research strategies
3. To familiarize students with the techniques of data collection, analysis of data and interpretation

SN	Course Outcomes	RBT Level
1	Understand research problem formulation.	L1, L2
2	Analyze research related information	L1, L2, L3, L4
3	Follow research ethics	L1, L2, L3
4	Understand that today's world is controlled by Computer, Information Technology, but tomorrow world will be ruled by ideas, concept, and creativity.	L1, L2, L3
5	Understanding that when IPR would take such important place in growth of individuals & nation, it is needless to emphasis the need of information about Intellectual Property Right to be promoted among students in general & engineering in particular.	L1, L2, L3, L4
6	Understand that IPR protection provides an incentive to inventors for further research work and investment in R & D, which leads to creation of new and better products, and in turn brings about, economic growth and social benefits.	L1, L2, L3, L4, L5

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction and Basic Research Concepts	07	L1, L2
	Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem, Need of Research in Business and Social Sciences. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations, Characteristics of Research: Systematic, Valid, Verifiable, Empirical and Critical.		
2	Types of Research	04	L1, L2, L3, L4, L5
	Basic Research, Applied Research, Descriptive Research, Analytical Research, Empirical Research. Effective literature studies approaches, analysis Plagiarism, Research ethics, Qualitative and Quantitative Approaches.		
3	Research Design and Sample Design	08	L2, L3, L4, L5
	Research Design – Meaning, Concepts, Types and Significance Sample Design – Meaning and Significance Essentials of a good sampling, Stages in Sample Design Sampling methods/techniques Sampling Errors & Non-Sampling Errors.		
4	Research Methodology	10	L2, L3, L4, L5
	Introduction, Meaning of Research Methodology, Stages in scientific Research Process, Identification, Selection & Formulation of Research Problem. Review of Literature, Formulation of Hypothesis, Formulation of Research Design, Sample Design, Data Collection – Data Analysis, Hypothesis testing & Interpretation of Data, Parametric tests, Non-parametric tests, Displaying of Data. Preparation of Research report.		
5	Formulating Research Problem	08	L2, L3, L4, L5
	Introduction, Research Problem Formulation mechanism, Considerations while formulating a Research Problem, Choice of Data – Primary Data Collection, Secondary Data Collection, Choice of Appropriate method for Data Collection. Analysis of Data, Interpretation of Analysis – Research Interpretation Techniques, Precautions of Interpretation.		

6	Outcomes of Research	08	
	Introduction, Research report writing, Effective technical writing, how to write report, types of report, Conclusions & Recommendations, Precautions while writing a Research Report, Validity Tests in Research, Ethical Issues in Research. Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee.		L2, L3, L4, L5
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Practical Research Methods	Dawson, Catherine	New Delhi, UBS Publishers Distributors	2nd	2002
2	Research Methodology-Methods and Techniques	Kothari, C.R.,	New Delhi, Wiley Eastern Limited	2nd	1985
3	Research Methodology-A Step-by-Step Guide for Beginners	Kumar, Ranjit,	Singapore, Pearson Education	2nd	2005

Online References:

Sr. No.	URL	Modules Covered
1	https://onlinecourses.nptel.ac.in/noc23_ge36/preview	M1, M2
2	https://archive.nptel.ac.in/courses/127/106/127106227/	M3, M4
3	https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/330	M1, M2, M3, M4, M5, M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Innovation & Entrepreneurship development and management								Course Code: OEC-7016		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	or/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: Foundational Business Knowledge, Innovation Fundamentals, Entrepreneurship Basics										

Course Objectives:

1. Gain a comprehensive understanding of innovation principles, models, and strategies to drive entrepreneurial success and enhance operational efficiency. To acquaint students with identifying problems for research and develop research strategies
2. Develop the skills to bridge innovation to entrepreneurship, leveraging organizational support, and implementing quality management methodologies for sustainable business growth.
3. Apply knowledge from these modules to make data-driven decisions, optimize processes, and achieve competitive advantage in dynamic business environments.

SN	Course Outcomes	RBT Level
1	Understand the fundamentals of innovation and its strategic importance in business.	L1, L2
2	Understand various innovation models and frameworks to enhance creative problem-solving.	L1, L2, L3
3	Understand how to transition innovative ideas into successful entrepreneurial ventures.	L1, L2, L3
4	Develop skills to provide support and resources to aspiring entrepreneurs within an organization.	L1, L2, L3, L4
5	Understand expertise in optimizing processes and resources to achieve efficient and effective operations.	L1, L2, L3, L4

6	Understand the principles of quality control and assurance to ensure product and service excellence.	L1, L2, L3, L4, L5
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Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction to Innovation and Innovation strategy	09	L1, L2
	Innovation: Innovations and their forms. Innovation, features and characteristics. Factors initiating innovations. Innovation process and its stages ,Innovation Strategy		
	Innovation Models		
2	Model of innovation. First-Generation Innovation Model (1G) – Technology Push, Second-Generation Innovation Model (2G)– Market Pull, Third-Generation Innovation Model (3G) – Coupling Method. Fourth-Generation Innovation Model (4G) – Integrated Model, Fifth-Generation Innovation Model(5G) – Network Model, Sixth-Generation Innovation Model (6G) – Open Innovation Model	08	L1, L2, L3, L4
3	Innovation to entrepreneurship	06	L1, L2, L3, L4, L5
	Entrepreneurial Motivation 2 Maslow’s theory, Herjburg’s theory ,McGragor’s Theory ,McClelland’s Need – Achievement Theory ,Culture & Society, Values / Ethics ,		
	Risk taking behavior		
	Organisation Assistance to Entrepreneur		
4	Assistance to an entrepreneur ,New Ventures, Industrial Park (Meaning, features, & examples) ,Special Economic Zone (Meaning, features & examples) , Financial assistance by different agencies , MSME Act Small Scale Industries , Carry on Business (COB) licence, Environmental Clearance , National Small Industries Corporation (NSIC) ,Government Stores Purchase scheme (e-tender process) , Excise exemptions and concession ,Exemption from income tax , Quality Standards with special reference to ISO ,Financial assistance to MSME , Modernisation assistance to small scale unit.	10	L1, L2, L3, L4, L5
5	Operations management		

	Production Forecasting, Planning And Control: Forecasting ,and Methods Life Cycle Analysis, Complete Forecasting , Production Planning , Production Planning as an integral part of the Corporate Planning Process ,Production Control, Integrative nature of Production Plans, Maintenance Management, Materials Management	12	L1, L2, L3, L4, L5
6	Quality Management	06	
	Quality-Concepts & tools, Cause and effect Diagram Control Chart , ISO 9000 System , Importance and need for Quality Systems , Management Responsibility ,Statistical Techniques (elementary ideas) ,Quality System Certification Procedure, Use case of start up		L1, L2, L3, L4, L5, L6
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Management, Operations Research, Industrial Engineering Business Management	Mahender Reddy Gaddam	PHI Learning Pvt. Ltd.	1st	2004
2	Production Planning and Control : A Comprehensive Approach	D R Kiran	McGraw-Hill Education	4th	2011
3	Sales Force Management: Leadership, Innovation, Technology	Mark W. Johnston and Greg W. Marshall	Routledge	12th	2020
4	Financial Technology: Case Studies in Fintech Innovation	Niels Pedersen	Palgrave Macmillan	1st	2018
5	Quality Concepts for the Process Industry	Michael Speegle	CRC Press	2nd	2011
6	Managing Quality: Concepts and Tasks	N.S. Sreenivasan	Pearson	5th	2008

Online References:

SN	URL	Modules
1	https://www.youtube.com/watch?v=rAB5EC1UHhc	M1, M2, M3, M4, M5, M6
2	https://www.youtube.com/watch?v=EXSwHz2GRV8	M1, M2, M3, M4, M5, M6
3	https://www.youtube.com/watch?v=yM9FitDaLv0	M1, M2, M3, M4, M5, M6

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic Student Development (CBCGS- HME 2023)
TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B.Tech. (All Branches)					SEM: VII					
Course Name: Intellectual Property Rights-Laws and practice					Course Code: OEC-7017					
Contact Hours Per Week: 03					Credits: 03					
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)		Practical / Oral (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	IA	IE	ESE	OR/PR	TW	Total
03	-	-	03	03	20	20	60	-	25	125
IA: In semester Assessment –Paper Duration - 1 Hours										
ESE: End Semester Examination – Paper Duration – 2 Hours										
Prerequisite: Intellectual Property and Business Law, Patenting.										

Course Objectives:

SN	Course Objectives	RBT Level
1	To introduce fundamental aspects of Intellectual property Rights to students who are going to play a major role in development and management of innovative projects in industries.	L1, L2, L3
2	To disseminate knowledge on various forms of intellectual property, including patents, copyrights, trademarks, and trade secrets, patent regime in India and abroad and registration aspects, patent drafting and searching.	L1, L2, L3
3	To familiarize students with key Indian intellectual property laws, such as the Patents Act, Copyright Act, and Trade Marks Act.	L1, L2, L3
4	To equip students with the knowledge of how to identify IP infringement and the remedies available under Indian law.	L1, L2, L3
5	To disseminate knowledge on Design, Geographical Indication (GI), Plant Variety and Layout Design Protection and their registration aspects.	L1, L2, L3, L4
6	To examine strategies for the commercialization of intellectual property, including licensing, technology transfer, and valuation, bioinformatics, and biotechnology advancements and to aware about current trends in IPR and Govt. steps in fostering IPR.	L1, L2, L3, L4

Course Outcomes:

SN	Course Outcomes	RBT Level
1	Students should be able to demonstrate a sound understanding of the fundamental concepts of intellectual property rights, including patents, copyrights, trademarks, and trade secrets, and their relevance to the engineering domain.	L1, L2, L3
2	Students should be capable of applying the principles and provisions of Indian intellectual property laws, such as the Patents Act, Copyright Act, and Trade Marks Act, to analyze, protect, and manage intellectual property in engineering projects.	L1, L2, L3
3	Students should be able to identify instances of IP infringement and evaluate the legal and practical implications of such infringements, including the ability to recognize and analyze prior art for patents.	L1, L2, L3
4	Students should be equipped with the skills required to enforce intellectual property rights and, if necessary, navigate IP litigation processes, including drafting cease and desist letters and understanding the roles of various stakeholders in IP enforcement.	L1, L2, L3
5	Students should be proficient in developing strategies for the commercialization of intellectual property, including drafting licensing agreements, conducting IP valuations, and managing an intellectual property portfolio effectively.	L1, L2, L3, L4
6	Students should have an awareness of emerging intellectual property issues and trends, such as the challenges and opportunities presented by digital intellectual property, bioinformatics, and biotechnology advancements.	L1, L2, L3, L4

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction to Intellectual Property Rights	08	L1, L2, L3
	1.1. Understanding Intellectual Property 1.2. Types of intellectual property (IP) 1.3. Importance of IP protection in engineering 1.4. Historical Development of IPR, Evolution of IPR laws in India 1.5. Key milestones in IPR history 1.6. International IP Treaties and Conventions 1.7. Overview of WIPO, TRIPS, and other international agreements 1.8. India's role in international IP protection		
2	Types of Intellectual Property	07	L1, L2, L3
	2.1. Copyright and Related Rights, Copyright basics Copyright protection in engineering works 2.2. Patents, Patent law and its application in engineering 2.3. Patentable subject matter and exclusions 2.4. Trademarks and Trade Secrets 2.5. Trademark registration and protection 2.6. Trade secret law and practices in India		
	Indian Intellectual Property Laws		

3	3.1. The Indian Patent Act 3.2. Patentable inventions under Indian law 3.3. Patent application and grant process 3.4. The Copyright Act 3.5. Copyright protection and registration in India, Fair use and exceptions 3.6. The Trade Marks Act 3.7. Trademark registration and enforcement 3.8. Geographical indications and certification marks	10	L1, L2, L3
4	IP Enforcement and Litigation 4.1. IP Infringement and Remedies 4.2. Identifying IP infringement 4.3. Civil and criminal remedies for IP violations 4.4. IP Enforcement Mechanisms 4.5. The role of police and customs in IP enforcement, Border control measures 4.6. IP Litigation in India, Overview of IP litigation process 4.7. Landmark IP cases in India	08	L1, L2, L3
5	IP Commercialization and Management 5.1. Licensing and Technology Transfer 5.2. Licensing agreements and their types 5.3. Technology transfer and commercialization 5.4. IP Valuation and Due Diligence 5.5. Valuation methods for intellectual property, Due diligence in IP transactions 5.6 IP Portfolio Management 5.7. Strategies for building and maintaining an IP portfolio, Cost-benefit analysis of IP protection	07	L1, L2, L3, L4
6	Emerging IP Issues 6.1 Digital Intellectual Property, IP in the digital age 6.2. Challenges and opportunities in digital IP 6.3. Bioinformatics and Biotechnology 6.4. IP considerations in biotech and bioinformatics 6.5. Recent developments in biotech IP	05	L1, L2, L3, L4

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Intellectual Property Rights: Protection and Management. India	Nithyananda, K V.	Cengage Learning India Private Limited		2019
2	Intellectual Property Rights. India	Neeraj, P., & Khusdeep, D	PHI learning Private Limited		2014
3	Law relating to Intellectual Property Rights	Ahuja, V K.	Lexis Nexis		2017
4	Intellectual Property Rights – An Overview	Subramanian, N., & Sundararaman, M..			2018
5	WIPO Intellectual property Handbook	World Intellectual Property Organisation.			2004

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic Student Development (CBCGS- HME 2023)
TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B.Tech. (All Branches)					SEM: VII					
Course Name: Contemporary Technology Development (Industry Case Study Based)					Course Code: OEC-7018					
Contact Hours Per Week: 03					Credits: 03					
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical / Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	IA	IE	ES E	PR	TW	Total
03	-	-	03	03	20	20	60	-	25	125
IA: In semester Assessment –Paper Duration - 1 Hours ESE: End Semester Examination – Paper Duration – 2 Hours										
Prerequisite: Basic Knowledge of Artificial Intelligence ,IoT, Industry 4.0, Renewable Energy Technology, Autonomous Vehicles and Green Technologies										

Course Objectives: To provide students with practical insights into the latest technological advancements across various industries. Through the analysis of real-world case studies, students will:

1. Understand the evolution and development of contemporary technologies.
2. Analyze the integration of these technologies into industries.
3. Explore challenges and innovations related to emerging technologies.
4. Develop the ability to assess technological solutions critically and propose improvements.

Course Outcomes:

SN	Course Outcomes	RBT Level
1	Gain an understanding of the most recent developments in technology across key industries.	L1,L2
2	Analyze how technological innovations are implemented and the impact on industrial processes.	L1,L2,L3
3	Apply knowledge to critically evaluate the effectiveness and future potential of new technologies.	L1,L2,L3
4	Develop a comprehensive understanding of how to manage technological transformations within industries.	L1,L2,L3,L4
5	Formulate strategies for leveraging emerging technologies to create competitive advantages in business.	L1,L2,L3,L4
6	Enhance critical thinking and analytical skills through case study analysis.	L1,L2,L3,L4,L5

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction to Contemporary Technology	08	L1, L2
	Overview of Technology Evolution, Key Sectors Influencing Technology Development such as Energy and Power, Healthcare and Biotechnology, Information Technology and AI, Manufacturing & Industry 4.0 and Sustainability and Green Technology		
2	Case Study 1 – Renewable Energy Technology	09	L1, L2, L3, L4
	● Technology Focus: Solar Energy, Wind Power, Energy Storage ● Industry Case: Development of Solar Farms in India ● Analysis: Challenges, Opportunities, and Future Growth Outcome: How renewable technologies are transforming global energy markets.		
3	Case Study 2 -Industry 4.0 and Smart Manufacturing	08	L1, L2, L3, L4, L5
	● Technology Focus: IoT, Automation, Additive Manufacturing, Data Analytics ● Industry Case: The Transformation of Automotive Manufacturing with IoT ● Analysis: Efficiency, Cost Reduction, and Sustainability Improvements Outcome: Evolution of manufacturing processes and predictive maintenance using smart technology.		
4	Case Study 3 – Artificial Intelligence and Healthcare	08	L1, L2, L3, L4, L5
	● Technology Focus: AI Algorithms, Predictive Analytics, Wearable Technologies ● Industry Case: AI in Predictive Diagnostics in Hospitals ● Analysis: Healthcare innovations, Data Security, Patient Outcomes Outcome: Assess how AI and machine learning are reshaping the healthcare industry.		
5	Case Study 4 – Autonomous Vehicles and Transportation	10	L1, L2, L3, L4, L5
	● Technology Focus: Autonomous Driving, EV Infrastructure ● Industry Case: Tesla's Development of Full Self-Driving Technology ● Analysis: Safety, Legal Challenges, Future of Mobility Outcome: The future of autonomous transport systems and infrastructure challenges.		
	Green Technologies and Sustainable Solutions		

6	- Technology Focus: Carbon Capture, Sustainable Energy Grids, Water Treatment Technologies - Industry Case: Development of Carbon Capture Solutions in the Oil Industry - Analysis: Environmental Impact, Policy Regulations, Long-Term Viability	10	L1, L2, L3, L4, L5, L6
	Outcome: Understanding the push towards sustainable technologies in heavy industries.		
	Total Hrs.	45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail.	Christensen, Clayton M.	Harvard Business Review Press	-	2016
2	Permissionless Innovation: The Continuing Case for Comprehensive Technological Freedom.	Thierer, Adam	Mercatus Center	-	2016
3	Industry report-The Future of Jobs Report	-	World Economic Forum	-	2023
4	The Industrial Internet of Things: Unleashing the Potential of Connected Products and Services.	-	McKinsey & Company	-	2022

Online References:

Sr. No.	URL	Modules Covered
1	MIT Technology Review – https://www.technologyreview.com	M1, M2, M3, M4, M5, M6
2	World Economic Forum – https://www.weforum.org	M1, M2, M3, M4, M5, M6
3	Industry 4.0 – https://www.industry40today.com	M1, M2, M3, M4, M5, M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Business Intelligence								Course Code: OEC-7019		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours The weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely completion of practical (40%) and Attendance / Learning Attitude (20%)										
Prerequisite: Basics of Computer Technology, Databases, algorithms										

Course Objectives: This course provides an overview of the technology of BI and the application of BI to an organization's strategies and goals.

SN	Course Outcomes	RBT Level
1	To understand the concepts of Business Intelligence	L1, L2, L3, L4
2	To interpret the terms related to Data warehouse	L1, L2, L3, L4
3	To conceptualize working of shingling in documents	L1, L2, L3, L4, L5
4	To illustrate the use of stream computing and understand concepts	L1, L2, L3, L4, L5
5	To acquaint with the realm of OLAP and OLTP	L1, L2, L3, L4, L5, L6
6	To acquaint with the different tools of BI and ethics in BI	L1, L2, L3, L4, L5, L6

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction To Business Intelligence	08	L1, L2, L3, L4
	Operational and Decision Support System, Data- Information-Knowledge-Decision making-Action cycle. Basic definitions- Business Intelligence; Data warehousing, Business Intelligence architecture, Use and benefits of Business Intelligence. Knowledge Discovery in Databases: KDD process model, Data Pre-processing: Cleaning: Missing Values; Noisy Values; Inconsistent values; redundant values. Outliers, Integration, transformation, reduction, Discretization: Equal Width Binning; Equal Depth Binning, Normalization, Smoothing.		
2	Introduction to Business Data Warehouse	07	L1, L2, L3, L4
	Definition of Data warehouse, Logical architecture of Data Warehouse, Data Warehouse model- Enterprise warehouse; Data Marts; Virtual warehouse. Populating business Data Warehousing: data integration and extract, transform, load (ETL).		
3	Shingling of documents	07	L1, L2, L3, L4, L5
	Finding Similar Items, Applications of Near-Neighbor Search, Jaccard similarity of sets, Similarity of documents, Collaborative filtering as a similar-sets problem, Documents, k Shingles, Choosing the Shingle Size, Hashing Shingles, Shingles built from Words. Similarity-Preserving Summaries of Sets, Locality-Sensitive hashing for documents. The Theory of Locality-Sensitive functions. Methods for high degrees of similarity.		
4	Introduction to Data Mining	07	L1, L2, L3, L4, L5
	Data mining definitions and process: business and data understanding. Association Analysis: Definition of association rule, General issues: Support; Confidence; Lift; Conviction, Frequent Item sets: APriori Algorithm; Issues with APriori Algorithm, Data structures: Hash tree and FP tree.		
5	Designing Business Data Warehouse	08	L1, L2, L3, L4, L5, L6
	OLTP and OLAP systems, Designing business information warehouse: Principles of dimensional modeling, Data cubes,		

	Data cube operations, data cube schema.		
6	Business Intelligence Tools and Technologies	08	
	Ethics in Business Intelligence, BI tools overview, Reporting and visualization tools, Principles of data visualization, Dashboard design best practices.		L1, L2, L3, L4, L5, L6
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Mining of Massive Datasets	Anand Rajaraman and Jeffrey David Ullman	Cambridge University Press	-	2012
2	Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses	Michael Minelli	Wiley	-	2013
3	Big Data for Dummies	J. Hurwitz, et al.	Wiley	-	2013
4	Understanding Big Data Analytics for Enterprise Class Hadoop and Streaming Data	Paul C. Zikopoulos, Chris Eaton, Dirk deRoos, Thomas Deutsch, George Lapis	McGraw-Hill	-	2012
5	Big data: The next frontier for innovation, competition, and productivity	James Manyika, Michael Chui, Brad Brown, Jacques Bughin, Richard Dobbs, Charles Roxburgh, Angela Hung Byers	McKinsey Global Institute	-	2011
6	Big Data Glossary	Pete Warden	O'Reilly	-	2011
7	Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph	David Loshin	Morgan Kaufmann Publishers	-	2013



Online References:

S. No.	Website Name	URL	Modules Covered
1	Tutorialspoint	https://www.tutorialspoint.com/what-is-business-intelligence	1-4
2	GeeksforGeeks	https://www.geeksforgeeks.org/clustering-in-machine-learning/	5
3	Simplilearn	https://www.simplilearn.com/tutorials/machine-learning-tutorial/classification-in-machine-learning	6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)					B.E. / B.Tech. SEM:VII					
Course Name: Supply Chain Management					Course Code: OEC-70110					
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	
3	-	-	3	3	20	20	60	-	25	125
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours The weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely completion of practical (40%) and Attendance / Learning Attitude (20%)										
Prerequisite: NIL										

Course Objectives: To acquaint with key drivers of supply chain performance and their inter-relationships with strategy and to study the complexity of inter-firm and intra-firm coordination in implementing programs such as e collaboration, quick response, jointly managed inventories, and strategic alliances.

SN	Course Outcomes	RBT Level
1	To acquaint with key drivers of supply chain performance and their interrelationships with strategy.	L1, L2, L3
2	To impart analytical and problem-solving skills necessary to develop solutions for a variety of supply chain management & design problems.	L1, L3, L4
3	To study the complexity of inter-firm and intra-firm coordination in implementing programs such as e-collaboration, quick response, jointly managed inventories and strategic alliances.	L1, L2, L3
4	To study the macro and micro dimensions and analyse techniques of logistics system	L2, L3, L4
5	To understand the role of transportation and evaluate the strengths and weaknesses of different modes of transportation.	L1, L3
6	To understand the importance of information and information technology in a supply chain.	L3, L4

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Building a Strategic Framework to Analyze Supply Chains	06	L1, L2, L3
	Supply chain stages and decision phases, Process view of supply chain: Supply chain flows, Examples of supply chains, Competitive and supply chain strategies, Achieving strategic fit: Expanding strategic scope, Drivers of supply chain performance. Framework for structuring drivers: inventory, transportation facilities, information obstacles to achieving fit.		
2	Designing the Supply Chain Network or Distribution Network	06	L1, L3, L4
	Distribution Networking: Role, Design, Supply Chain Network (SCN): Role, Factors, Framework for design decisions. Parameters to decide suitable distribution		
3	Materials Management	07	L1, L2, L3
	Scope, Importance, Classification of materials, Procurement, Purchasing policies, Vendor development and evaluation. Inventory control systems of stock replenishment, Cost elements, EOQ and its derivative modules.		
4	Dimensions of Logistics	08	L2, L3, L4
	Introduction: A Macro and Micro Dimensions, Logistics interfaces with other areas, Approach to analyzing logistics system, Logistics and systems analyzing: Techniques of logistics system analysis, factors affecting the cost and Importance of logistics.		
5	Warehouse and Transport Management	08	L1, L3
	Concept of strategic storage, Warehouse functionality, Warehouse operating principles, Developing warehouse resources, Material handling and packaging in warehouses, Transportation Management, Transport functionality and principles, Transport infrastructure, transport economics and Pricing. Transport decision making.		
6	IT in Supply Chain		

	6.1 IT framework, Customer Relationship Management (CRM), internal Supply chain management, Supplier Relationship Management (SRM) and Transaction Management. Coordination in a Supply Chain 6.2 Lack of supply chain coordination and the Bullwhip effect, Obstacle to Coordination, Managerial levers, Building partnerships and trust. Emerging Trends and Issues 6.3 Vendor managed inventory-3PL-4PL, Reverse logistics: Reasons, Role, Activities; RFID systems: Components, Applications, Implementation; Lean supply chain, Implementation of Six Sigma in supply chain, Green supply chain.	10	L3, L4
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Supply Chain Management Strategy, Planning, and operations	Sunil Chopra and Peter Meindl	Pearson	6th Edition	2016
2	Designing & Managing Supply chain	David Simchi Levi, Philip Kaminsky & Edith Smichi Levi	McGraw Hill	3rd Edition	2007
3	Supply Chain Redesign: Transforming Supply Chains into Integrated Value Systems	Robert B Handfield, Ernest L Nicholas	Prentice Hall	-	2002
4	The Management of Business Logistics: A Supply Chain Perspective	Coyle, Bardi, Langley	Thomson learning	-	2003
5	Supply chain management: for global competitiveness	B S Sahay	Macmillan	-	1999

Online References:

S. No.	Website Name	URL	Modules Covered
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TCET
DEPARTMENT OF INFORMATION TECHNOLOGY (IT)
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education
Under Autonomy - CBCGS-HME 2023
University of Mumbai



1	https://nptel.ac.in	https://nptel.ac.in/courses/110/106/110106045/	2
2	https://nptel.ac.in	https://nptel.ac.in/courses/110/107/110107074/	3
3	https://www.scmhub.com	https://www.scmhub.com/courses/BBA	2
	https://www.udemy.com	https://www.udemy.com/topic/supply-chain/	4

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)							B.E. / B.Tech. SEM:VII		
Course Name: Digital Marketing and E-commerce							Course Code: OEC-70111		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)				
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation				
Hours Per Week					Theory (100)		Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	ESE	OR/PR	TW	125
3	-	-	3	3	40	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours ESE: End Semester Examination - Paper Duration - 2 Hours									
Prerequisite: Discrete Mathematics									

Course Objectives:

1. Understand the Fundamentals of Digital Marketing
2. Develop E-commerce Skills and Knowledge
3. Apply Data Analytics and Metrics in Decision-Making
4. Integrate Engineering and Technology into Marketing Efforts
5. Hands-On Experience with Digital Marketing Tools

SN	Course Outcomes	RBT Level
1	Understand the basic concepts Digital Marketing and E-commerce	L1, L2
2	Understand and apply Digital Marketing Fundamentals for Website Development and Optimization	L1, L2
3	Understand , create E-commerce Strategies, Platforms and Tools	L1, L2, L3
4	Social Media Marketing	L1, L2, L3
5	Online Advertising and Analytics	L1, L2, L3, L4
6	E-commerce Marketing Strategies	L1, L2, L3, L4, L5

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction to Digital Marketing and E-commerce	07	L1, L2
	Overview of Digital Marketing and E-commerce, Evolution and Importance of Digital Marketing, Key E-commerce Concepts and Trends, Digital Marketing vs. Traditional Marketing, Creating a Digital Marketing Strategy, Content Creation and Marketing		
2	Digital Marketing Fundamentals	08	L1, L2, L3, L4, L5
	Website Development and Optimization, Exploring E-commerce Platforms (e.g., Shopify, WooCommerce, Magento), Setting Up an Online Store, Payment Gateways and Security, Inventory Management and Fulfillment, Search Engine Optimization (SEO), Pay-Per-Click Advertising (PPC), Email Marketing, Social Media Marketing, Content Marketing		
3	E-commerce Platforms and Tools	08	L2, L3, L4, L5
	E-commerce Business Models, Building and Managing an Online Store, Customer Relationship Management (CRM), E-commerce Payment Systems, Security and Trust in E-commerce		
4	Social Media Marketing	08	L2, L3, L4, L5
	Social Media Platforms and their Usage, Creating Effective Social Media Content, Social Media Advertising, Social Media Analytics and Insights		
5	Online Advertising and Analytics	07	L2, L3, L4, L5
	Display Advertising, Remarketing and Retargeting, Web Analytics Tools (e.g., Google Analytics), Key Metrics and KPIs, A/B Testing and Conversion Rate Optimization (CRO)		
6	E-commerce Marketing Strategies	07	L2, L3, L4, L5, L6
	E-commerce SEO and SEM, Customer Acquisition and Retention Strategies, Mobile Commerce (M-commerce), Influencer Marketing in E-commerce, Emerging Trends in Digital Marketing and E-commerce		
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	"Digital Marketing for Dummies	Ryan Deiss and Russ Henneberry	For Dummies	1st Edition	2017
2	The Art of SEO	Eric Enge, Stephan Spencer, and Jessie Stricchiola	O'Reilly Media	3rd Edition	2015
3	Contagious: How to Build Word of Mouth in the Digital Age	Jonah Berger	Simon & Schuster	Reprint edition	2016
4	E-commerce Evolved: The Essential Playbook To Build, Grow & Scale A Successful E-commerce Business"	Tanner Larsson	CreateSpace Independent Publishing Platform	1st edition	2016
5	Influence: The Psychology of Persuasion	Robert B. Cialdini	HarperCollins	Revised edition	2006

Online References:

SN	Website Name	URL	Modules
1	Google Digital Garage	https://learndigital.withgoogle.com/digitalgarage	Module 1, 2, 3, 4, 5, 6
2	HubSpot Academy	https://academy.hubspot.com/	Module 1, 2, 3, 4, 5, 6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Industrial Safety and Management								Course Code: OEC-70112		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: Environmental Science, Basic Engineering Principles, Statistics and Data Analysis										

Course Objectives: These course aim to provide a well-rounded understanding of industrial safety and equip students with practical skills to ensure a safe and sustainable industrial environment. These course also emphasize the importance of leadership, communication, and ethical considerations in the field of Industrial Safety and Management.

SN	Course Outcomes	RBT Level
1	Students will have a comprehensive understanding of industrial safety principles, including risk assessment, hazard identification.	L1, L2, L4
2	Students will be able to develop and implement effective Safety Management Systems (SMS) for industrial settings, encompassing safety policies, planning, documentation, training, and emergency response protocols.	L2, L4, L5
3	Students will possess the skills to identify various workplace hazards, both physical and chemical, and implement control measures to mitigate these hazards, with a focus on engineering-specific scenarios.	L2, L4, L5
4	Students will gain expertise in the selection, proper use, maintenance, and inspection of Personal Protective Equipment (PPE) relevant to industrial and engineering environments.	L1, L4, L6
5	Students will be capable of demonstrating leadership in safety practices, effectively communicating safety policies, conducting safety meetings, and promoting a safety-conscious culture within industrial teams.	L5, L6, L3

6	Students will develop an understanding of environmental safety regulations, sustainable practices, and corporate social responsibility related to industrial operations, contributing to responsible and sustainable industrial practices.	L4, L5, L6
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Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction to Industrial Safety and Management	08	L1, L2, L4
	Importance of Industrial Safety in Engineering, Historical Perspective of Industrial Accidents, Occupational Safety and Health Act (OSHA) Overview, Safety Culture in Engineering Industries, Risk Assessment and Hazard identification, Accident Investigation and Reporting, Safety Regulations and Compliance for Engineers		
2	Safety Management Systems	08	L2, L4, L5
	Introduction to Safety Management Systems (SMS) in Engineering, Elements of an Effective SMS, Safety Policy and Objectives, Safety Planning and Documentation, Safety Training and Communication, Safety Audits and Inspections, Emergency Response and Preparedness		
3	Engineering Hazards and Controls	08	L2, L3, L5
	Physical Hazards in Engineering: Machinery and Equipment Safety, Chemical Hazards in Engineering: Hazardous Material Handling, Biological Hazards in Engineering: Control Measures, Ergonomic Hazards in Engineering: Workspace Design, Electrical Safety in Engineering, Fire Safety and Prevention in Engineering, Confined Space Entry in Engineering Projects		
4	Personal Protective Equipment (PPE)	05	L1, L4, L6
	Introduction to PPE for Engineers, Selection and Use of PPE for Engineering Tasks, Maintenance and Inspection of PPE, Respiratory Protection, Hearing Protection, Eye and Face Protection, Hand and Foot Protection, Fall Protection		
5	Safety Leadership and Communication in Engineering	08	L5, L6, L3
	Role of Leadership in Safety for Engineering Teams, Safety Committees and Employee Involvement in Engineering, Effective Safety Communication, Behavioral Safety and Motivation, Safety Meetings and Toolbox Talks for Engineering Teams, Safety Training Techniques for Engineers, Conflict Resolution and Safety in Engineering Environments		

6	Environmental Safety and Sustainability in Engineering	08	L4, L5, L6
	Environmental Regulations and Compliance, Hazardous Waste Management in Engineering, Sustainable Practices in Projects, Energy Efficiency and Conservation in Engineering, Green Technology and Safety for Engineers		
	Pollution Prevention, Environmental Impact Assessment, Corporate Social Responsibility and Safety in Engineering		
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Introduction to Industrial and Systems Engineering	Wayne C. Turner and Steve M. Easterling	Pearson	2 nd Edition	1987
2	Occupational Health and Safety Management: A Practical Approach	Charles D. Reese	CRC Press	1 st Edition	2017
3	Safety and Health for Engineers	Roger L. Brauer	Wiley	3 rd Edition	2016
4	Industrial Safety and Health Management	C. Ray Asfahl and David W. Rieske	Prentice Hall	7 th Edition	2018
5	Handbook of OSHA Construction Safety and Health	Charles D. Reese and James Vernon Eidson	CRC Press	2 nd Edition	2006

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Renewable Energy Technologies								Course Code: OEC-70113		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: Thermodynamics										

Course Objectives: Course intend to provide working principles of various renewable energy sources and their utilities

SN	Course Outcomes	RBT Level
1	Demonstrate need of different renewable energy sources	L1, L2
2	Discuss importance of renewable energy sources	L1
3	Discuss various renewable energy sources in Indian context	L1, L2, L3, L4
4	Calculate and analyse utilization of solar and wind energy	L1, L2, L3, L4
5	Illustrate design of biogas plant	L1, L2, L3, L4
6	Demonstrate basics of hydrogen energy	L1, L2

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
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1	Introduction to Energy Sources	07	L1, L2
	Renewable and non-renewable energy sources, Need for Renewable Energy Sources, Energy Consumption as a measure of Nation's development; Strategy for meeting the future energy requirements, Global and National scenarios,		
	Prospects of renewable energy sources, Present status and current installations, Introduction to Hybrid Energy Systems.		
2	Solar energy	12	L1
	Merits and demerits, Solar radiation - beam and diffuse radiation, solar constant, earth sun angles, attenuation and measurement of solar radiation, local solar time, derived solar angles, sunrise, sunset and day length, Methods of Solar Radiation estimation. Solar Energy collection devices and Classification: Flat plate collectors, concentrating collectors, Solar air heaters- types, solar driers, storage of solar energy-thermal storage, solar pond , solar water heaters, solar distillation, solar still, solar cooker, solar heating & cooling of buildings, Solar Photovoltaic systems & applications.		
3	Wind Energy	10	L1, L2, L3, L4
	Principle of wind energy conversion; Basic components of wind energy conversion systems; wind mill components, various types and their constructional features; design considerations of horizontal and vertical axis wind machines: analysis of Aerodynamic forces acting on wind mill blades and estimation of power output; wind data and site selection considerations.		
4	Energy from biomass	06	L1, L2, L3, L4
	Biomass conversion technologies, Biogas generation plants, classification, advantages and disadvantages, constructional details, site selection, digester design consideration, filling a digester for starting, maintaining biogas production, Fuel properties of bio gas, utilization of biogas.		
5	Geothermal Energy		

	Estimation and nature of geothermal energy, geothermal sources and resources like hydrothermal, geo-pressured hot dry rock, magma. Advantages, disadvantages and application of geothermal energy, prospects of geothermal energy in India. Energy from the ocean: Ocean Thermal Electric Conversion (OTEC) systems like open cycle, closed cycle, Hybrid cycle, prospects of OTEC in India. Energy from tides, basic principle of tidal power, single basin and double basin tidal power plants, advantages, limitation and scope of tidal energy. Wave energy and power from wave, wave energy conversion devices, advantages and disadvantages of wave energy	08	L1, L2, L3, L4
6	Hydrogen Energy	05	L1, L2
	Methods of Hydrogen production, Hydrogen Storage, Fuel Cells and Types of Fuel Cells.		
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Non-conventional energy sources	G.D. Rai	Khanna Publishers	1 st Edition	2004
2	Renewable Energy: Power for a Sustainable Future	Edited by Godfrey Boyle	Oxford University Press	3 rd Edition	2006
3	Solar Energy: Principles of Thermal Collection and Storage	SP Sukhatme and J K Nayak	TMH	2nd Edition	2004
4	Wind and Solar Power Systems	Mukund R Patel	CRC Press	1 st Edition	2006

Online References:

S. No.	Website Name	URL	Modules Covered
1	www.nptel.ac.in	https://nptel.ac.in	M1-M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Sustainable Agriculture								Course Code: OEC-70114		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: N/A										

Course Objectives: This course aims to explore the principles and practices of sustainable agriculture, with a focus on environmentally friendly, socially responsible, and economically viable approaches to food production.

SN	Course Outcomes	RBT Level
1	Understand the principles and goals of sustainable agriculture.	L1, L2, L3
2	Analyse the environmental, social, and economic impacts of conventional agriculture.	L1, L2, L3, L4
3	Explore alternative farming practices and their benefits.	L1, L2, L3
4	Develop practical skills for implementing sustainable farming techniques.	L1, L2, L3, L4
5	Evaluate the role of policy and technology in sustainable agriculture.	L1, L2, L3, L4
6	Understanding of sustainable farming techniques and their broader societal impacts.	L1, L2, L3, L4

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
	Introduction to Sustainable Agriculture		

1	Definition and principles of sustainable agriculture, Historical perspectives and development of sustainable farming, Environmental and social challenges in modern agriculture, The role of agroecology in sustainable agriculture, Ecosystem services provided by sustainable farming systems	07	L1, L2
2	Organic Farming, Soil Health and Conservation Principles and practices of organic agriculture, Certification and labelling of organic products, Challenges and opportunities in organic farming, Importance of soil health in sustainable agriculture, Soil conservation techniques, Soil testing and nutrient management	08	L1, L2
3	Crop Rotation and Diversification, Integrated Pest Management (IPM) Benefits of crop rotation and diversification, Planning and implementing crop rotation systems, Polyculture and intercropping, IPM principles and strategies Biological, chemical, and cultural control methods, Pest monitoring and decision-making in IPM	08	L1, L2, L3
4	Sustainable Livestock Farming, Water Management and Irrigation Principles of sustainable animal agriculture, Pasture management and rotational grazing, Animal welfare and ethical considerations, Efficient water use in agriculture, Sustainable irrigation practices, Water quality and conservation	08	L1, L2, L3
5	Sustainable Farming Technologies Emerging technologies in sustainable agriculture, Precision agriculture and data-driven farming, Genetic modification and biotechnology, Government policies and incentives for sustainable farming, Economic viability of sustainable agriculture	08	L1, L2, L3
6	Sustainable Agriculture and Food Security The role of sustainable agriculture in ensuring food security, Global and local food systems, Challenges and opportunities in feeding a growing population	06	L1, L2, L3
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Sustainable Agriculture and Agronomy	Farrell Waltz	CALLISTO REFERENCE	Illustrated edition	2019
2	Soil Science For Regenerative Agriculture	Amélie des Plantes	Independently published	1 st edition	2023
3	Crop Rotation and Cover Cropping	Seth Kroeck	Chelsea Green	Updated edition	2020
4	Artificial Intelligence and IoT-Based Technologies for Sustainable Farming and Smart Agriculture	Pradeep Tomar, Gurjit Kaur	Engineering Science Reference	3rd edition	2021

Online References:

S. No.	Website Name	URL	Modules Covered
1	www.nature.com	https://www.nature.com/articles/s41893-020-0510-0	M1 – M2
2	www.ceew.in	https://www.ceew.in/publications/sustainable-agriculture-india	M3 - M5
3	www.coursera.org	https://www.coursera.org/learn/sustainable-agriculture	M3 – M6

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Fundamentals of Disaster Management								Course Code: OEC-70115		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: N/A										

Course Objectives: Main objective of the subject is to understand causes of different types of disasters, mitigation /rehabilitation measures and existing government policies and agencies.

SN	Course Outcomes	RBT Level
1	Get to know natural as well as manmade disaster and their extent and possible effects on the economy.	L1, L2, L3
2	Plan of national importance structures based upon the previous history.	L1, L2, L3
3	Get acquainted with government policies, acts and various organizational structure associated	L1, L2, L3
4	Get to know the simple do's and don'ts in such extreme events and act accordingly.	L1, L2, L3
5	Understand role of Government and Non government agencies in financing the relief measures	L1, L2, L3
6	Understanding of government policies and procedures for mitigation of disasters	L1, L2, L3

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
1	Introduction	04	Understand (U)
	Definition of Disaster, hazard, global and Indian scenario, general perspective, importance of study in human life, Direct and indirect effects of disasters, long term effects of disasters. Introduction to global warming and climate change		
2	Natural Disaster and Manmade disasters	09	
	Natural Disaster: Meaning and nature of natural disaster, Flood, Flash flood, drought, cloud burst, Earthquake, Landslides, Avalanches, Volcanic eruptions, Mudflow, Cyclone, Storm, Storm Surge, climate change, global warming, sea level rise, ozone depletion, Manmade Disasters: Chemical, Industrial, Nuclear and Fire Hazards. Role of growing population and subsequent industrialization, urbanization and changing lifestyle of human beings in frequent occurrences of manmade disasters		Understand (U)
3	Disaster Management, Policy and Administration	09	Understand (U)
	Disaster management: meaning, concept, importance, objective of disaster management policy, disaster risks in India, Paradigm shift in disaster management Policy and administration: Importance and principles of disaster management policies, command and co-ordination of in disaster management, rescue operations-how to start with and how to proceed in due course of time, study of flowchart showing the entire process.		
4	Institutional Framework for Disaster Management in India	07	Understand (U)
	Importance of public awareness, Preparation and execution of emergency management program. Scope and responsibilities of National Institute of Disaster Management (NIDM) and National disaster management authority (NDMA) in India. Methods and measures to avoid disasters, Management of casualties, set up of emergency facilities, importance of effective communication amongst different agencies in such situations. Use of Internet and softwares for effective disaster management. Applications of GIS, Remote sensing and GPS in this regard.		
5	Financing Relief Measures		

	Ways to raise finance for relief expenditure, role of government agencies and NGO's in this process, Legal aspects related to finance raising as well as overall management of disasters. Various NGO's and the works they have carried out in the past on the occurrence of various disasters, Ways to approach these teams. International relief aid agencies and their role in extreme events	10	Understand (U)
6	Preventive and Mitigation Measures	06	Understand (U)
	Pre-disaster, during disaster and post-disaster measures in some events in general. Structural mapping: Risk mapping, assessment and analysis, sea walls and embankments, Bio shield, shelters, early warning and communication. Non		
	Structural Mitigation: Community based disaster preparedness, risk transfer and risk financing, capacity development and training, awareness and education, contingency plans. Do's and don'ts in case of disasters and effective implementation of relief aids.		
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Disaster Management	Harsh K.Gupta	Universities Press Publications	-	2003
2	Disaster Management: An Appraisal of Institutional Mechanisms in India	O.S.Dagur	Centre for land warfare studies	-	2011
3	Introduction to International Disaster Management	Damon Copolla	Butterworth Heinemann Elsevier Publications	-	2006
4	Disaster Management Handbook	Jack Pinkowski	CRC Press Taylor and Francis group	-	2008
5	Disaster management & rehabilitation	Rajdeep Dasgupta	Mittal Publications	-	2007
6	Natural Hazards and Disaster Management, Vulnerability and Mitigation	R B Singh	Rawat Publications	-	2006
7	Concepts and Techniques of GIS	C.P. Lo Albert, K.W. Yonng	Prentice Hall (India) Publications.	-	2006

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: Waste Management and Energy Recovery								Course Code: OEC-70116		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: N/A										

Course Objectives: The objective is to equip students with knowledge of sustainable waste management practices and energy recovery technologies, evaluating economic and environmental impacts while addressing relevant policies and real-world challenges.

SN	Course Outcomes	RBT Level
1	To learn broader understandings on various aspects of waste management practiced in industries.	L1, L2
2	To learn recovery of products from solid waste to compost and biogas, incineration and energy recovery,	L1, L2
3	To learn hazardous waste management and treatment, and integrated waste management.	L1, L2,L3
4	Interpret technologies for generation of energy from solid waste	L1, L2
5	Select thermochemical conversion methods	L1, L2
6	Identify sources of energy from bio-chemical conversion	L1, L2, L3

Detailed Syllabus:

Module No.	Topics	Hrs.	RBT Levels
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1	INTRODUCTION TO WASTE MANAGEMENT:	07	L1, L2
	Classification of wastes (source and type based), waste management (SWM), elements of SWM, ESSWM (environmentally sound waste management) and EST (environmentally sound technologies), factors affecting SWM, Indian scenario, progress in MSW (municipal solid waste) management in India. Indian and global scenario of e-waste,		
2	WASTE GENERATION ASPECTS:	08	L1, L2
	Waste stream assessment (WSA), waste generation and composition, waste characteristics (physical and chemical), health and environmental effects (public health and environmental), comparative assessment of waste generation and composition of developing and developed nations, a case study results from an Indian city, handouts on solid waste compositions. E-waste generation.		
3	COLLECTION, STORAGE, TRANSPORT AND DISPOSAL OF WASTES:	08	L1, L2, L3
	Waste Collection, Storage and Transport: Collection components, storage-containers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system, a case study. Waste Disposal: key issues in waste disposal, disposal options and selection criteria, sanitary landfill, landfill gas emission, leachate formation, environmental effects of landfill, landfill operation issues, a case study.		
4	WASTE PROCESSING TECHNIQUES & SOURCE REDUCTION, PRODUCT RECOVERY & RECYCLING:	08	L1, L2
	Purpose of processing, mechanical volume and size reduction, component separation, drying and dewatering. Source Reduction, Product Recovery and Recycling: basics, purpose, implementation monitoring and evaluation of source reduction, significance of recycling, planning of a recycling program, recycling program elements, commonly recycled materials and processes, a case study		
5	THERMOCHEMICAL CONVERSION:		

	Incineration, pyrolysis, gasification of waste using gasifiers, environmental and health impacts of incineration; strategies for reducing environmental impacts. Energy production from wastes through incineration, energy production through gasification of wastes. Energy production through pyrolysis and gasification of wastes, syngas utilization.	07	L1, L2
6	BIO-CHEMICAL CONVERSION & ENERGY PRODUCTION FROM WASTE PLASTICS,GAS CLEAN UP WASTE, HEAT RECOVERY:		
	Anaerobic digestion of sewage and municipal wastes, direct combustion of MSW-refuse derived solid fuel, industrial waste, agro residues, anaerobic digestion biogas production, and present status of technologies for conversion of waste into energy, design of waste to energy plants for cities, small townships and villages. Energy production from wastes through fermentation and trans esterification. Cultivation of algal biomass from wastewater and energy production from algae. Energy production from organic wastes through anaerobic digestion and fermentation, introduction to microbial fuel cells. Process analysis and reactor configurations for Methane production, Energy assessment, Bio-methanation from sludge digestion, Concept of conversion efficiency, energy waste, waste heat recovery classification, advantages and applications, commercially viable waste heat recovery devices.	07	L1, L2, L3
Total Hours		45	

Books and References:

SN	Title	Authors	Publisher	Edition	Year
1	Integrated Solid Waste Management,	Tchobaanoglous, G., Theisen, H., and Samuel A Vigil,	McGraw-Hill Publishers	-	1993.
2	Waste Management	Bilitewski B., Hard He G., Marek K., Weissbach A., and Boeddicker H	Springer	-	1994

3	Thermo-chemical Processing of Biomass: Conversion into Fuels, Chemicals and Power,	Robert C. Brown,	John Wiley and Sons, USA,	-	2019
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B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)					B.E. / B.Tech. SEM:VII					
Course Name: German Language I Elementary use of language (A1 & A2)					Course Code: OEC-70117					
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	
3	-	-	3	3	20	20	60	-	25	125

Certification Exam (A1 & A2): Gothe

LEVEL A1

Learning Objective:

Can understand and use familiar, everyday expressions and very simple sentences, which relate to the satisfying of concrete needs.

Can introduce him/herself and others as well as ask others about themselves – e.g. where they live, who they know and what they own – and can respond to questions of this nature.

Can communicate in a simple manner if the person they are speaking to speaks slowly and clearly and is willing to help.

Curriculum:

Grammar	Conversation	Vocabulary
● Conjugation Of Regular Verbs	● Pronunciation Rules	● The Alphabet
● The Nominative	● How To Introduce Yourself	● Colors
● Numbers	● How To Make Appointments	● Number
● Gender Rules	● Using Public Transport	● Family

● Plural	● Looking For Apartments	● Body
● Irregular Verbs	● Asking For / Giving Directions	● Clothing
● Formal Salutation	● How To Say The Time And Date	● At home
● Word Order Of Main Clauses	● Doctor's Visit	● Food & Beverages
● Separable And Inseparable Verbs	● How To Order Food In A Restaurant	● Animals
● Modal Verbs		● Professions
● Negation		● Weather
● How To Form Questions		● Leisure Time Activities
● Conjunctions		● Emotions & Adjectives
● The imperative		● Countries & Nations
		● Means of Transport

LEVEL A2

Learning Objective:

Can understand sentences and commonly used expressions associated with topics directly related to his/her direct circumstances (e.g. personal information or information about his/her family, shopping, work, immediate surroundings).

Can make him/herself understood in simple, routine situations dealing with a simple and direct exchange of information on familiar and common topics.

Can describe his/her background and education, immediate surroundings and other things associated with immediate needs in a simple way.

Curriculum:

● Grammar:	● Conversation:
✓ The perfect tense	✓ Professions and activities
✓ Verbs with dative	✓ Telling about the past
✓ Verbs with accusative	✓ Daily schedule leisure activities
✓ Main and subordinate clauses	✓ In the office, on the phone
✓ Subjunctive II	✓ Vacation
✓ Nouns indicators of time	✓ Living: garden, city etc.
✓ Verbs + Prepositions	✓ Technology
✓ Prepositions of time	✓ Environment
✓ Future Tense	✓ Health

✓ Modal verbs in the past tense	✓ Making appointments
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Topics to learn:

- Introducing oneself in German: Name, Country, Place of residence, Languages
- Greetings - Different times of the day
- Asking others information about them: Name, Country, Place of Residence, Languages
- Numerals: 0 to 20 - Asking and telling phone/mobile number
- Asking and responding to “How are you?”
- Formal and informal forms for addressing another person
- Introducing oneself and asking information in German: Hobbies and Profession
- Yes-No questions
- Introducing other person
- The German Alphabet - Number from 20 onward
- Nationalities and languages
- Days of the week
- Giving additional information about work (where, what position, when)
- Months of the year
- Seasons
- The German article (definite)
- Places in a city
- Landscapes in and around a city
- Modes of transport
- The German article (indefinite and negation)
- Asking and giving directions
- The Imperative
- Food, beverages and daily need items
- Meal times in the day
- Where do you buy what?



- When do you eat what?
- The Verb position
- The accusative case
- Shopping at a supermarket

Reference Link for Certification: <https://www.goethe.de/ins/in/en/spr/kon/stu.html>

B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2025-26)

B.E. / B. Tech. (All Branches)								B.E. / B.Tech. SEM:VII		
Course Name: French Language I								Course Code: OEC-70118		
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical / Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR/PR	TW	125
3	-	-	3	3	20	20	60	-	25	
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination										

The six French language certification levels are A1, A2, B1, B2, C1, and C2:

- A1: Beginner or breakthrough level
- A2: Elementary or waystage level
- B1: Intermediate or threshold level
- B2: Upper intermediate or vantage level
- C1: Advanced or effective operational proficiency level
- C2: Proficiency or mastery level
- **Reference Link for Certification:** <https://www.goethe.de/ins/in/en/spr/kon/stu.html>

DELF (LEVEL A1): BEGINNER OR BREAKTHROUGH LEVEL

A1.1 Listening	A1.2 Listening
• Can recognize his/her name when it is spoken by another person. • Can understand basic greetings, farewells, and common expressions of politeness (e.g. hello, goodbye, sorry). • Can recognize and understand the words for people around them (e.g. teacher, mother, brother, doctor). • Can often recognize spoken words similar to the languages with which they are familiar (e.g. international, vacation, accident, work). • Can understand simple questions which directly concern them such as their name and their address. • Can understand simple instructions, such as “Stand up, please”, “Come here”, or “Close the door, please” when expressed slowly and clearly, possibly with accompanying gestures and pictures. • Can understand simple information such as name, age, and perhaps country of origin, when that information is given slowly and clearly. • Can understand simple information about an object (for example the size and colour of a backpack, who it belongs to and where it is) if the speaker speaks slowly and clearly, possibly with accompanying gestures. • Can recognize words, names, and numbers previously learned when heard in a short, simple recording delivered at a slow pace. • Can identify the sound of most letters heard in context. • Can identify the names of common colours when heard.	• Can understand what is being said provided people speak slowly and carefully and there are significant pauses. • Can understand numbers, prices, times, and other pieces of short information if given slowly and clearly. • Can understand and follow basic instructions from peers (e.g., for catching a bus, for playing a basketball game in the gym). • Can recognize basic phrases that denote position (under, over, beside) and can understand simple instructions including directions (e.g. on the second floor, near the bus stop, third door on the left, etc.). • Can understand some words and expressions when others are talking about themselves, their family, or their classroom provided the speech is clear and slow. • Can understand what is being said about the colour, size, and ownership of computers, cell phones, houses, etc. if information is spoken clearly and slowly. ✓ Can understand the most important information in a straightforward talk provided something is already known about the subject and that the talk is accompanied by pictures or drawings. ✓ Can understand most of a story when it is read slowly and clearly and is accompanied by pictures or drawings. ✓ Can understand simple calculations using ‘plus’, ‘minus’, and ‘times’. ✓ Can recognize numbers and shapes when heard as well as some mathematical terms. ✓ Can recognize isolated vocabulary and terms from specific subject areas. ✓ Can recognize the names of some sports and the accompanying equipment. ✓ Can listen to instructions for games and follow teacher’s or students’ modeling of the activity.

A1.1 Spoken Interaction	A1.2 Spoken Interaction
• Can provide his/her name and ask others for theirs. • Can point to an object and ask what it is. • Can express a lack of understanding. • Can use basic expressions of greeting, farewell, and politeness (e.g. hello, goodbye, please and thank you). • Can understand and respond in a predictable pattern to simple questions about familiar things (e.g. name, age, and birthplace) if the other person speaks slowly and clearly. • Can ask for something (in a store for example) when pointing or gesturing supports the request. • Can make an appointment with someone (for example the doctor's office) • Can talk about vacation and past experiences using basic expressions • Can talk about future plans using basic expressions • Can use basic expressions to talk about the weather	• Can interact in a simple way, provided others are prepared to repeat, rephrase, and speak slowly. • Can answer simple questions using individual words, expressions, or short sentences. • Can ask the location of a familiar object. Can also answer this type of question if asked clearly. • Can make simple needs understood (e.g. ask for the loan of a pencil or other classroom material). Students are also able to offer these items to others. • Can express how they are feeling and also inquire as to how others feel. • Can express the size and name the colour of familiar objects. Can also ask others about these qualities. • Can meet and take leave of people using appropriate expressions. • Can make introductions of self and others. • Can communicate the day, date, and time as well as ask about those matters. ☐ Can indicate a lack of comprehension. ☐ Can ask for assistance with vocabulary and clarification of task. ☐ Can use learned phrases, questions, and responses.
A1.1 Spoken Production	A1.2 Spoken Production
• Can express how they are feeling using simple, standard expressions. • Can name some common objects in familiar environments (e.g. items in the house and commonplace food items). • Can express basic personal information such as name, age, address, and country of origin. • Can use a very limited number of grammatical constructions (e.g. "My name is...").	• Can introduce themselves very briefly, for example stating name, where they are from, and what school they attend. • Can provide basic information about familiar things (e.g. favourite subjects; description of their home or classroom; number, ages, and occupations of family members). • Can identify the main body parts. • Can express preferences. • Can briefly talk about weekend or holiday plans. ☐ Can name some subject specific vocabulary terms. Can ☐ express likes/dislikes about a familiar topic.
A1.1 Reading	A1.2 Reading

• Can understand brief, simple instructions if encountered previously in the same or similar form. • Can recognize some words and straightforward sentences in simple texts if seen previously. • Can recognize the main information (e.g. date, price, time) on posters, brochures, signs, and invitations. • Can recognize some expressions and straightforward sentences in simple texts if allowed to use a dictionary.	• Can understand words and expressions on signs when everyday language is used (e.g. 'Open/Closed', 'Exit'). • Can adequately understand straightforward forms in order to insert personal information such as name, address, date of birth, etc. • Can recognize important information on basic promotional material (e.g. price, date, and time on posters). • Can recognize short messages and greetings (e.g. "back in an hour"; "happy birthday"). ☐ Can slowly read simple, short text sentence by sentence and pick out the main information. ☐ Can recognize previously encountered parts of words, prefixes, and suffixes. ☐ Can identify vocabulary related to the topic. ☐ Can connect labels on diagrams with appropriate pieces of science equipment. ☐ Can identify key words (e.g. names of people and places) in text. ☐ Can access main ideas of text when accompanied by illustrations. ☐ Can read familiar names of sports, games, and players.
A1.1 Writing	A1.2 Writing
• Can copy or write own name. • Can write simple words and very simple sentences with assistance like a dictionary or textbook. • Can copy or write labels on familiar objects in a picture or diagram (e.g. boy, girl, chair, house, etc.). • Can write straightforward information about themselves in short sentences or provide that information on a questionnaire.	• Can write very simple informal messages (e.g. a text message or a note to tell someone when and where to meet). • Can write brief messages like post cards and birthday greetings. • Can write simple information about themselves (e.g. name, age, address, and hobbies). • Can write simple descriptions of everyday objects (e.g. colour and size of a car; a brief description of their home). • Can fill in a questionnaire with personal details. ☐ Can complete gapped sentences using a word list of familiar words. ☐ Can label charts, equipment, and diagrams.
A1. Grammar Focus	A1. Vocabulary Focus
Articles defined, indefinite, etc • The interrogation, negation • The gender of nouns, adjectives • Possessive and demonstrative adjectives, • Some prepositions, The reflexive verbs • The present, the past tense, the near future, imperative, future, imperfect • pronouns COD (the, they) and COI (him, her), pronouns (I, you, he, she, we, they, them)	Numbers, nationalities, occupations, family, daily activities, colours, clothing, furniture, personal things, types of building, places, where places are located, stores, transportation, food, celebrations, housing, vacation, the media, weather

DELF (LEVEL A2): ELEMENTARY LEVEL

A2.1 Listening	A2.2 Listening
• Can understand the main points in short, simple stories and reports when there is some previous understanding of the topic. • Can understand the main points in simple, clear, and short audio announcements and messages. • Can understand short, clear and simple instructions and explanations when delivered slowly. • Can understand others' expression of feelings (e.g. whether they are indicating they are happy/sad; well/ill; etc.). • Can extract important information from audio texts (e.g. concert announcements, sports scores, weather forecasts, etc.) provided the message is delivered clearly. ☐ Can understand the important points of a story or account if it is told slowly and clearly. ☐ Can understand the most important points in a straightforward talk or presentation accompanied with drawings and/or diagrams provided there was some previous familiarity with the topic. ☐ Can understand isolated, familiar words and phrases when listening to clear, slow, and basic text read aloud. ☐ Can understand phrases and high frequency vocabulary related to areas of personal relevance.	• Can generally comprehend the gist of overheard conversations. • Can generally identify changes of topic in conversation. • Can understand short, straightforward audio messages if these messages are delivered slowly and clearly (e.g. an announcement of a bus delay, an intercom announcement at school, a news bulletin, etc.). • Can understand an audio advertisement if it is delivered clearly and concerns a product and/or service of interest to the student. • Can understand the dialogue of a movie, a trailer for a film, or the events in a news story when the visuals provide contextual support. ☐ Can understand simple technical explanations if given slowly and clearly and opportunity is given for clarification (e.g. how to use different types of equipment). ☐ Can understand a presentation on a topic when the subject is familiar to them and it is delivered slowly and clearly (e.g. another student's report on a classroom project, an interview with a person of a similar age exploring common interests). ☐ Can listen to other students' contributions in class, and use these responses as models for their own. ☐ Can get the main idea of a short text when read aloud clearly and slowly. ☐ Can get the main idea of audio-visual or other media presentations on familiar topics. ☐ Can follow verbal instructions (e.g. for playing a game, for using a cash dispenser, for completing a science procedure, for constructing a time line, etc.) if the instructions are spoken clearly and slowly. ☐ Can follow text in a book when it is being read aloud slowly and clearly.
A2.1 Spoken Interaction	A2.2 Spoken Interaction

• Can express common greetings such as, “happy birthday”, “happy new year”. • Can express common feelings (e.g. I am cold; I am hungry, I am thirsty, etc.). • Can participate in a short telephone conversation with some preparation in advance. • Can ask straightforward questions in familiar situations (e.g. asking the location and price of an item, asking for directions, asking for the time, etc.) and understand the responses. • Offer others different things to eat or drink. • Can exchange information about everyday matters using simple vocabulary (e.g. weekend activities, hobbies and other interest, shopping, etc.). ☐ Can ask for assistance in understanding textbook problems. ☐ Can ask for clarification of unknown words and terms. ☐ Can ask how to express something in another language (e.g. “How do you say ‘kilómetro’ in English?”).	• Can ask for a word to be spelled out and can spell out names or words themselves. • Can ask appropriate questions during a conversation to ensure the other person understands points being made or information being given. • Can ask someone to say something more clearly, to explain something a different way, or to repeat what has been said. • Can take part in simple social interactions asking people what they do at work/school and what they do in their spare time and can reply to these sorts of questions when asked. • Can ask for information (e.g. how to get to a location) as well as offer similar, simple explanations to others. ✓ Can ask other students to confirm solutions to problems. ☐ Can ask for assistance (e.g. understanding textbook problems, asking for clarification of vocabulary, terminology, rules etc.).
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A2.1 Spoken Production	A2.2 Spoken Production
• Can indicate personal strengths and weaknesses. • Can describe their home, how to get there and the length of time spent in travelling there. • Can describe basic aspects of their day by day life such as favourite foods and daily activities. • Can talk briefly about familiar topics such as weekend or holiday plans. ✓ Can talk about a familiar topic in a short presentation, prepared in advance, but not read.	Can describe, using simple words, a daily event at home or school. • Can describe, using simple words, their opinion of a situation or event. • Can provide a simple description of a common object (e.g. describe the appearance and behaviour of an animal with which they are familiar). • Can express what they like and/or dislike about something with which they are familiar (e.g. a song, a book, a particular place, etc.). Can provide some brief indication of reasons for their opinions. • Can provide a short, simple account of something experienced (e.g. a class trip, a party, a shopping outing). • Can provide some information concerning their culture’s celebrations and special events. ☐ Can produce a summary of a short story expressed in a simple sequence of events. ☐ Can say most numbers (including years) without

	hesitation. ☐ Can describe simple mathematical or scientific processes (e.g. Then add the two numbers together and divide by four.”. “Fill the beaker with water.”). ☐ Can talk about familiar places, historical figures, and events using simple vocabulary and language constructions.
A2.1 Reading	A2.2 Reading
• Can understand most of what is written in short, simple texts on subjects with which they are familiar and/or in which they are interested. • Can locate important information in simple text (e.g. dates and times on posters; ‘keep to the right’ or ‘no parking’ signs; people’s ages and places of residence) in newspaper or magazine articles. • Can understand the gist of short articles and ads in magazines (print and/or electronic) provided the subjects are familiar. ☐ Can understand most of what occurs in a well-structured short story and can identify the story’s main characters. ☐ Can read and understand texts written in simple language using common words and/or words recognized from other known languages. ☐ Can recognize some key terminology from subject areas (e.g. labels on science equipment and the legend on maps). ☐ Can follow a set of clear-cut instructions (e.g. a simple recipe, a set of directions), especially if there are pictures or diagrams to illustrate the most important steps or if the procedure has been previously demonstrated.	• Can distinguish and understand the important information in everyday visual text (e.g. the weather forecast, advertising material, local brochures, and school newsletters). • Can follow instructions for using everyday equipment (e.g. cash dispensers, vending machines, ticket dispensers). • Can understand short instructions illustrated by step by step visuals (e.g. cooking instructions on food packages; instructions for mobile phone use, and simple science experiments). • Can use everyday reference materials like catalogues or yellow pages. • Can understand the main information in short articles and reports (e.g. discussions on the Internet, letters to the editor, a list of school rules) if they deal with familiar subjects. • Can understand the main points in a straightforward personal communication if the topic is familiar. ✓ Can read patterned and predictable texts. ☐ Can extract important information, especially relating to facts, from textbooks and other text. ☐ Can understand information set out in mathematical language. (Will still need assistance with decoding of problems written in everyday language). ☐ Can read diagrams, with accompanying text, and use information to answer questions if provided with lead-in phrases. ☐ Can read, using a bilingual dictionary, short narratives with some illustrations as support. ☐ Can read short text segments to find main ideas and a few details. ☐ Can identify relevant section of textbook or other information source.

A2.1 Writing	A2.2 Writing
• Can write, either by hand or electronically, simple notes of greeting, good wishes, appreciation, and thanks. • Can write, either by hand or electronically, an informal invitation to others their age, reply to such an invitation, or confirm/cancel/change an invitation. • Can write short sentences expressing wants and preferences. • Can write, with the help of illustrations, an explanation of where they live and how to get there. • Can use simple sentences and expressions to describe such things as their surroundings, their daily activities and the people around them. • Can prepare simple reminders or notes for personal use (e.g. diary, journal, homework book, etc.). ☐ Can write an introduction or conclusion to a story with the help of a dictionary. ☐ Can copy dates and facts from short, simple text. ☐ Can label charts, diagrams, and maps.	• Can provide information on everyday issues (e.g. will be late for school, will attend an event, is ill, etc.). • Can provide descriptions of common objects or familiar places using short, simple sentences. • Can describe an aspect of daily life, an event, or a personal experience very briefly in simple language (e.g. a hobby, a party, a celebration, an accident). Can include concrete details such as what, where, and when. • Can exchange useful information with friends in e-mails or short letters. • Can complete a questionnaire providing background information of personal skills, interests, and education. ☐ Can tell a simple story (e.g. the best holiday ever or life in the future). ☐ Can use simple grammatical structures accurately. ☐ Can write straightforward text on topics familiar to them. Can link these accounts with transitional words such as 'and', 'but' and 'because'. Can use important connecting words to indicate chronological order (e.g. 'then', 'after', 'later', etc.). ☐ Can write a summary sentence of a text's main points giving one or two details. ☐ Can give personal reactions to a piece of age-appropriate literature. ☐ Can use text as a stimulus to write short dialogue. ☐ Can write simple phrases to describe an experiment when guided by a writing frame. ☐ Can produce illustrations with short descriptions for a school or class project. ☐ Can write about class work or lab work in a learning journal when guided by a writing frame. ☐ Can devise a time line for a topic. ☐ Can fill in missing facts, names, and subject specific terms in gapped texts. ☐ Can write notes with the aid of a writing frame (e.g. word web or headings). ☐ Can write down the key words during an oral presentation.



B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2024-25)

B.E. / B. Tech. (All Branches)					B.E. / B.Tech. SEM:VII					
Course Name: Japanese Language I					Course Code: OEC-70119					
Teaching Scheme (Program Specific)					Examination Scheme (Formative / Summative)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)			Practical/ Oral (25)	Term Work (25)	Total
Theor y	Tutorial	Practical	Contact Hours	Credit s	ISE	IE	E S E	OR/PR	TW	
3	-	-	3	3	20	20	60	-	25	125
ISE: In-Semester Examination - Paper Duration – 1 Hours IE: Innovative Examination ESE: End Semester Examination - Paper Duration - 2 Hours										
Prerequisite: Hiragana and Katakana										

Course Objectives: To build foundational proficiency in basic Japanese language skills, including the ability to read and write Hiragana, Katakana, and 100 kanji, comprehend and participate in simple spoken conversations, and demonstrate an understanding of essential cultural norms for everyday communication.

Course Outcomes:

SN	Course Outcomes	RBT Level
1	Mastery of Basic Vocabulary and Kanji	L1, L2
2	Proficiency in Japanese Script	L1, L2, L3
3	Understanding Basic Grammar and Sentence Structure	L1, L2, L3, L4
4	Listening Comprehension in Everyday Situations	L1, L2, L3, L4

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
Module 1	Japanese Writing Systems Objective: Learn and master Hiragana and Katakana.		L1, L2, L3

	Introduction to the Japanese writing system (Hiragana and Katakana) Learn and practice reading/writing Hiragana characters Learn and practice reading/writing Katakana characters Exercises to read simple words in Hiragana/Katakana	05	
Module 2	Basic Greetings and Self-Introductions Objective: Engage in basic conversations and introductions. Basic greetings (おはよう, こんにちは,こんばんは, etc.) Polite forms (です, ます) Introducing oneself (name, country, occupation) Practice short dialogues in self-introductions Simple listening exercises	05	L1, L2
Module 3	Numbers, Time, and Counters Objective: Use numbers, tell time, and understand basic counters. Numbers (1–100) and larger numbers Telling time, dates, and days of the week Basic counters for objects (個, 本, 枚, etc.) Asking and answering questions about time and quantity Listening and speaking practice	05	L1, L2, L3, L4
Module 4	Basic Grammar and Sentence Structure Objective: Understand sentence structure and basic particles.		

	Japanese sentence structure (Subject-Object-Verb) Particles (は, が, を, に, で, etc.) Forming simple sentences (affirmative, negative, past, present) Introduction to verbs in ます-form (present tense) Writing and speaking practice using basic sentences	05	L1, L2, L3, L4
Module 5	Verbs and Adjectives (5 hours) Objective: Use common verbs and adjectives in sentences. Common verbs (食べる, 飲む, 行く, 来る, etc.) Adjectives: i-adjectives and na-adjectives (おいしい, きれい, etc.) Sentence structures using adjectives (describing objects, people) Practicing conjugation of ます-form verbs Exercises in listening, speaking, and writing	05	L1, L2, L3, L4
Module 6	Expressing Existence and Locations Objective: Talk about existence and locations. Verbs of existence: いる (for living things) and ある (for non-living things) Location words (上, 下, 中, 前, 後, そば, etc.) Asking and answering “where” questions Describing the location of objects and people Dialogue and comprehension practice	05	L1, L2, L3, L4, L5

Books and References:

Sr. No.	Name of Book	Author Name	Edition
1	"Genki I: An Integrated Course in Elementary Japanese"	Eri Banno, Yutaka Ohno, Yoko Sakane, Chikako Shinagawa, Kyoko Tokashiki	3rd Edition (Revised)
2	"Minna no Nihongo I"	3A Corporation	2nd Edition (Revised)
3	"Japanese for Busy People I"	Seiichi Makino, Michio Tsutsui	Revised 4th Edition
4	JLPT N5 Official Practice Workbook	Japan Educational Exchanges and Services (JEES)	2020 Edition
5	Kanji Look and Learn	Eri Banno, Yutaka Ohno, Yoko Sakane, Chikako Shinagawa	2nd Edition
6	Shin Nihongo 500 Mon N5		2020

Online References:

Sr. No	Website Name	URL	Modules Covered
1	JLPT Official Website	https://www.jlpt.jp/e/	1
2	Wani Kani (Kanji Learning)	https://www.wanikani.com/	2
3	NHK World Easy Japanese	https://www.nhk.or.jp/lesson/english/	3
4	Jisho.org (Online Japanese Dictionary)	https://jisho.org/	4
5	Anki (Spaced Repetition Software)	https://apps.ankiweb.net/	5
6	Bunpro (Grammar Practice)	https://www.bunpro.jp/	6



B.E. Semester –VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2024-25)

B.E. (Information Technology)							S.E. SEM: VII		
Course Name: Privacy and Security in Online Social Media & Linux for Sys-Ads							Course Code: SP-IT-IS-701		
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)				
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation				
Hours Per Week : 3					Theory (100)		Practical/Or al (25)	Ter m Wor k (25)	Total
Theor y	Tutoria l	Practica l	Contac t Hours	Credit s	ISE 1	ISE 2	PR/OR	TW	125
3	-	-	3	3	50	50	--	-	
ESE: End Semester Examination - Paper Duration - 3 Hours									
Prerequisite: Basic / Intermediate programming course.									

Course Objective: Privacy and security of online social media need to be investigated, studied and characterized from various perspectives (computational, cultural, psychological, etc.). Student completing the course will be able to appreciate various privacy and security concerns (spam, phishing, fraud nodes, identity theft) on Online Social Media and Student will be able to clearly articulate one or two concerns comprehensively on one Online Social Media. Module 4 and 5 aims to provide understanding regarding linux OS and administrative commands of linux.

Detailed Syllabus:

Module No.	Topics	NPTEL/MOOC Link
1	Online Social networks & data gathering	https://onlinecourses.nptel.ac.in/noc21_cs28/preview
	What is Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs , Collecting data from Online Social Media.	
2	Trust & Reputation	https://onlinecourses.nptel.ac.in/noc21_cs28/preview
	Trust, credibility, and reputations in social systems	

3	Policies & Privacies Online social Media and Policing , Information privacy disclosure, revelation and its effects in OSM and online social networks	https://onlinecourses.nptel.ac.in/noc21_cs28/preview
4	Phishing Phishing in OSM & Identifying fraudulent entities in online social networks, Refresher for all topics	https://onlinecourses.nptel.ac.in/noc21_cs28/preview
5	Linux Installation & Basic commands Linux File system, Installation , Basic file handling commands	https://onlinecourses.swayam2.ac.in/aic20_sp26/preview
6	Linux administrative commands Creating user account , assigning group on user creation, user password management, modifying user account, group password and login , administrating group with gpasswd command	https://onlinecourses.swayam2.ac.in/aic20_sp26/preview

Total Hours:- 40 (12 Weeks)

Syllabus for Blockchain Specialization –A.Y. (2024-25)

B.E. Semester –VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023) TCET

Autonomy Scheme (w.e.f. A.Y. 2024-25)

B.E. (Information Technology)					B.E. SEM: VII				
Course Name: Blockchain and Hyperledger Fabric					Course Code: SP-IT-BT701				
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)				
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation				
Hours Per Week					Theory (100)	Practical/ Oral (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hour s	Cred its	ISE	ESE	PR/OR	TW	100
3	-	2	5	4	40	60	-	-	
ISE: In- Semester Examination(In campus online in form of MCQs) - Paper Duration – 1 Hour ESE: End Semester Examination (Subjective in offline mode) - Paper Duration - 2 Hours									
Prerequisite: basic fundamentals of computers, c programming languages									

Course Objective: This subject intends to deliver the fundamental design and architectural primitives of Blockchain, the system and the security aspects, along with various use cases from different application domains.

Detailed Syllabus:

Module No.	Topics
1	Topic Name: Blockchain Architecture
	Contents: Introduction to Blockchain – I (Basics, History, Architecture, Conceptualization) ,Basic Crypto Primitives, Bitcoin Basics Distributed Consensus
2	Topic Name: Permissioned Blockchain
	Contents: Consensus in Bitcoin – I (The Basics, PoW and Beyond, The Miners) Permissioned Blockchain (Basics, Consensus), Permissioned Blockchain(RAFT Consensus, Byzantine General Problem, Practical Byzantine Fault Tolerance) Blockchain for Enterprise – Overview ,Blockchain Components and Concepts

3	Topic Name: Hyperledger Fabric
	Contents: Hyperledger Fabric – Transaction Flow ,Hyperledger Fabric Details Fabric – Membership and Identity Management,Hyperledger Fabric Network Setup,Fabric Demo on IBM Blockchain Cloud ,Hyperledger composer
4	Topic Name: Blockchain Applications
	Contents: Blockchain in Financial Service (Payments and Secure Trading, Compliance and Mortgage, Financial Trade),Revolutionizing Global Trade Blockchain in Supply Chain, Blockchain in Other Industries, Blockchain in Government (Advantages, Use Cases, Digital Identity), Blockchain in Government(Hyperledger Indy, Tax Payments and Land Registry Records) Blockchain Security (Overview, Membership and Access control in Fabric,Privacy in Fabric)
5	Topic Name: Blockchain Security
	Contents: Blockchain Security(Fabric SideDB)
	Research Aspects(Consensus Scalability, Bitcoin-NG, Collective Signing, Byzcoin), Research Aspects(Algorand,Cross Fault Tolerance, Secured Multi-Party Computation) Blockchain for Science (Blockchain for Big Data, Blockchain and AI)
6	Topic Name Comparing Ecosystems
	Contents Comparing Ecosystems – Ethereum,Comparing Ecosystems – Ethereum development tools and Quorum, Comparing Ecosystems – Corda

Total Hours:- 40 (12 Weeks)

Capstone Project hours:

Work to be done	Hrs.
Identification and Study of blockchain application	4
Project Title Identification	2
Modelling or prototype design	4
Graphics Design	4
Implementation	8
Testing of Mini Project	4
Preparation of Report	4
Total Hours	30

Online Resources:

Sr. No.	Website Name	URL	Modules covered
1.	https://www.coursera.org	https://www.coursera.org/learn/blockchain-in-platforms	M1,
2	https://www.coursera.org	https://www.coursera.org/learn/blockchain-platforms	M2,
3	https://www.coursera.org	https://www.coursera.org/lecture/blockchain-platforms/hyperledger-part-1-87Qz4	M3
4	https://www.coursera.org	https://www.coursera.org/learn/blockchain-platforms	,M4
5	https://www.coursera.org	https://www.coursera.org/learn/blockchain-platforms	,M5,
6	https://www.coursera.org	https://www.coursera.org/learn/blockchain-platforms	M6



B.E. Semester – VII

Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS- HME 2023)

TCET Autonomy Scheme (w.e.f. A.Y. 2024-25)

BE (Information Technology)							SEM: VII		
Course Name: Industry Certification							Course Code: PROJ-IT701		
Contact Hours Per Week: 04							Credits: 02		
Teaching Scheme (Program Specific)					Examination Scheme (Formative/ Summative)				
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation				
Hours Per Week					Theory (100)		Practical/Oral (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	IA	ESE	PR	TW	50
-	-	4	4	2	-	-	25	25	
The weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely completion of Practical (40%) and Attendance (20%)									

.# Students are required to do industry certification from the selected specialization